Transportation Research Part C 152 (2023) 104150

Contents lists available at ScienceDirect

Transportation Research Part C

journal homepage: www.elsevier.com/locate/trc

ADMM-based joint rescheduling method for high-speed railway timetabling and platforming in case of uncertain perturbation ☆

Xuan Liu [a], Min Zhou [a,*], Hairong Dong [a], Xingtang Wu [a], Yidong Li [b], Fei-Yue Wang [c]

[a] State Key Laboratory of Rail Traffic Control and Safety, Beijing Jiaotong University, Beijing 100044, China
[b] School of Computer and Information Technology, Beijing Jiaotong University, Beijing 100044, China
[c] State Key Laboratory of Management and Control for Complex Systems, Institute of Automation, Chinese Academy of Sciences, Beijing 100190, China

ARTICLE INFO

Keywords:
Timetable rescheduling
Platforming
Uncertain perturbation
Model predictive control
Alternative direction method of multipliers

ABSTRACT

The operation of high-speed trains is susceptible to perturbations such as strong winds and heavy snowfall, causing them to deviate from operating schedules. A joint mixed-integer linear programming (MILP) model based on the event-activity network is proposed to reschedule the train timetable and platform assignment jointly. Trains traveling in different directions are simultaneously considered in this model, and they can occupy the available reverse platforms in stations. Furthermore, to strengthen the robustness of the rescheduling scheme in case of uncertain perturbations, such as variable temporary speed restrictions caused by changing wind speed, the uncertainty is modeled using scenario-based chance constraints, which are then converted into deterministic constraints. Model predictive control (MPC) divides the entire time horizon into several stages. In each stage, the proposed model is decomposed according to directions. The constraints for trains traveling in different directions are dualized by the Alternating Direction Method of Multipliers (ADMM), which makes the sub-problems decoupled and solvable in parallel in each iteration of ADMM, promoting real-time performance. The real timetable and line data of Beijing–Shanghai HSR are utilized to investigate the effect of the proposed method. Compared to the commonly used strategies, it can significantly reduce delays and the number of affected events. The proposed method also shows relatively high robustness compared to the method in the case of predefined perturbations. It causes fewer conflicts and fewer needs for rescheduling when facing the changing perturbation and gets rescheduling results with higher quality.

1. Introduction

As the coverage of High-speed Railways (HSRs) expands, the operating environment of high-speed trains has become increasingly complex, leading to higher requirements for punctuality. However, perturbations such as bad weather, equipment failures, and large passenger flows have become more frequent in recent years, causing significant delays that can spread quickly between trains and stations due to increasing traffic density and short headways. If effective adjustment measures are not taken, delays can seriously affect the operation efficiency of HSRs and passenger satisfaction (Dong et al., 2021). When perturbations occur, train dispatchers must reschedule the timetables, which is a time-consuming and challenging task that requires them to provide a safe and

☆ This article belongs to the Virtual Special Issue on IG005585: VSI:Advanced railway transpo.

* Corresponding author.
E-mail address: zhmin@bjtu.edu.cn (M. Zhou).

https://doi.org/10.1016/j.trc.2023.104150
Received 15 June 2022; Received in revised form 18 March 2023; Accepted 21 April 2023
Available online 23 May 2023
0968-090X/© 2023 Elsevier Ltd. All rights reserved.

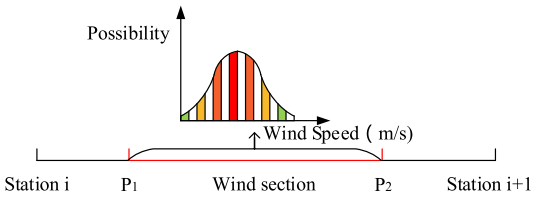


Fig. 1. Frequently changing wind speed.

conflict-free adjustment strategy to minimize delays and quickly resume normal operations. In practice, when perturbation occurs, train dispatchers typically start by calling drivers and asking trains in affected sections to decelerate. Then, dispatchers release the temporary speed restriction orders to all the following trains. After that, they start to reschedule and release the adjusted operation plan. They always start with several trains and then reschedule the following trains repeatedly until the delay is eliminated or the operation ends. This process is time-consuming, especially for high-congested HSRs. The commonly used approach is moving the operation lines of the following trains parallel. However, this approach sacrifices significant efficiency and wastes the railway infrastructure. To maintain punctuality and minimize delays in the face of perturbations such as bad weather, equipment failures, large passenger flow, or matter intrusion, train dispatchers must quickly reschedule the timetables of high-speed trains. This involves adjusting the arrival and departure times of trains and, in severe situations, breaking the original departure order, changing routes and platforms, or even canceling part of trains. The challenging and time-consuming process requires dispatchers to adjust timetables using limited railway infrastructure resources rationally. Therefore, generating a conflict-free rescheduling scheme and quickly eliminating delays are critical for HSRs.

Generating a conflict-free rescheduling scheme and quickly eliminating delays are critical for HSRs. The rescheduling scheme involves not only timetabling but also platform assignment at stations. Platform assignment is just as important in disruption as the train timetable. The timetables specify the running time in sections and the dwell time at stations, while platform assignment can guide how trains travel within stations. Together, these components make up an integral rescheduling plan. Usually, the train timetable is rescheduled first, followed by the platform assignment, which can be generated based on the adjusted timetable or even ignored. However, this sequential(only timetabling) optimization method sometimes fails to fully meet actual requirements. On the one hand, the train timetabling and platform assignment are tightly coupled. Although sequential optimization eliminates the coupling between these two problems, it cannot guarantee that the solution of the platform assignment for a given station, when constrained to the arrival and departure times found by timetabling problems, always exists (Zhang et al., 2020b). On the other hand, this kind of rescheduling also neglects the internal structure of the stations in detail, which results in low efficiency in using railway infrastructures.

In addition, perturbations, such as strong winds, are highly uncertain. The wind speed frequently changes, as illustrated in Fig. 1. For safe operations, trains must decelerate below the restricted speed when strong wind occurs, resulting in additional running time in the affected sections. As a result, trains must take longer to run in the affected sections. Fig. 1 shows the frequently changing wind speed. The wind section, marked by the red line, begins from point $P1$ and ends at point $P2$. Because the wind speed frequently changes, and thus the additional running time in the affected section is also changeable. The uncertainty of changeable wind speed can render the adjustment result infeasible if only a predefined wind speed is considered. Meanwhile, the robustness of the rescheduling result is relatively low, which cannot handle the randomness of perturbation. Frequent adjustments are required once the state of the perturbation changes. When making adjustments, determining the additional running time is very complicated. The most common method is estimating additional running time to replace all uncertain scenarios. This method causes the adjustment strategies to be ineffective or cause unsafe operations. The uncertainty of perturbations has relatively high requirements for real-time robustness of adjustment strategy.

This paper aims to address the challenge of severe speed restrictions resulting from the uncertainty caused by strong winds, with the primary objective of jointly optimizing the timetables and platform assignments of high-speed trains under such uncertain perturbations. To achieve this goal, we employ several strategies, including retiming, reordering, rerouting, and additional dwell.

When a train has a scheduled dwell in a station, it must dwell after rescheduling; otherwise, it can either pass through the station or dwell there as originally scheduled. Moreover, trains are allowed to occupy the reverse routes and platforms. For instance, downstream trains can use the routes and platforms initially available for upstream trains.

For the sake of simplification, the station is considered in detail. The routes and platforms within stations are considered. Furthermore, the interlocking mechanism used in this paper is 'Route-Lock-Route-Released(RLRR)', which means that a route can only be occupied after another route has been cleared for a certain time, to avoid conflicts between routes. However, the sections between stations is simplified by ignoring the blocks and signals. Safe trains following processes is ensured by guaranteeing that the departure headway in stations and the arrival headway in the following stations are greater than the minimum values based on the interlocking mode. This simplification reduces the number of constraints in the model while still retaining the critical constraints necessary to avoid train conflicts in stations.

A joint optimization model is proposed in this paper to solve the research gaps in joint rescheduling for train timetabling and platforming under uncertain perturbations, which, in this case, are multiple long-duration and strong winds. The influences of perturbations are regarded as the additional running time in sections where perturbations happen. In order to ensure the safe

operation of trains in stations, the safe headway between conflict routes is guaranteed based on the interlocking mechanism "RLRR", which can avoid conflicts and reduce the number of decision variables and constraints. Previous methods treated perturbations as constant, and trains in different directions were solved independently, which resulted in decreased robustness, feasibility, and quality. The proposed method aims to minimize total arrival delay and simultaneously obtain a robust, re-optimized timetable and reasonable platform assignment that meets actual requirements. The main contributions of this paper are given as follows:

1. An event-activity network-based train rescheduling model with retiming, reordering, additional dwell, and rerouting that reschedules train timetable with considering platforming is proposed. It considers the inbound routes, platforms, and outbound routes of stations in detail. The model allows trains in different directions to occupy the reverse routes and platforms in stations. This model uses railway resources as much as possible to speed up the elimination of delays based on satisfying the safe operation of trains.
2. The additional running time in the section caused by perturbation is regarded as a stochastic parameter. An uncertain perturbation is divided into several possible scenarios to deal with the uncertain perturbation. The scenarios are generated according to the classical wind speed prediction function, weber probability distribution function. Each scenario has a deterministic additional running time with a probability of occurrence. Scenario-based chance constraints involved in these scenarios are constructed and linearized to deterministic constraints.
3. The entire time horizon is divided into many stages by utilizing MPC to reduce the complexity and scale of the problem. In each stage, the model is decomposed by ADMM. The constraints about trains in different directions are dualized, and the original problem is decoupled into sub-problems. The sub-problems are computed in parallel in each iteration of ADMM. The proposed method is named ADMM-J in the paper.
4. The proposed method is tested on the Beijing–Shanghai HSR with several degrees of speed restriction and uncertainty.

The remainder of this paper is organized as follows. Section 2 gives a comprehensive literature review on train rescheduling, joint optimization for timetabling and platforming, and rescheduling with uncertainty. Section 3 analyzes the problem and gives the assumptions. A joint optimization model for the train timetable and platform assignment is given in Section 4. In Section 5, the solution methods, MPC, and ADMM are introduced. In Section 6, a case study based on actual data evaluates the effectiveness of the proposed method. Finally, the conclusion and future work are discussed in Section 7.

2. Related work

In recent years, real-time train rescheduling has attracted the attention of many researchers and has been a research hotspot (Cacchiani et al., 2014). In general, perturbations can be divided into two categories based on their severity: disturbances, which are minor perturbations caused by slight speed restrictions, disruptions, which are severe perturbations resulting from blockages and significant speed restrictions (Xu et al., 2017a). The impact of these perturbations is far-reaching and can lead to train cancellations. Accordingly, train rescheduling research can be divided into three categories: Train rescheduling in the case of disturbances, disruptions, or both. According to the severity of the perturbations, different strategies are employed for train rescheduling. In train rescheduling in case of disturbance, train timetabling with a short duration under the temporary speed restriction is the primary problem. Generally, this typically involves adjusting the arrival and departure times of trains. Thus, retiming and skipping(adding) stops are the most commonly used strategies. Sometimes, reordering is also considered. If a disruption occurs, longer durations and a wider range of influence necessitate more complex strategies such as reordering, rerouting, and train cancellations. Short-turning is also an efficient strategy in some countries, especially for frequent operations (Zhu and Goverde, 2019).

2.1. Train timetable rescheduling problems

D'Ariano (2009) developed a decision system, Railway traffic Optimization through Alternative graphs (ROMA), to cope with real-time train rescheduling in case of disturbances. The blocking time theory and signaling were modeled by the alternative graph (AG). The built-in algorithm of the system can solve the traffic control problem. Corman et al. (2010) proposed a tabu search algorithm with retiming and rerouting strategies. Furthermore, they incorporated it into ROMA. The calculation experiment in train delay and track blockage showed that this heuristic algorithm performs well. To improve the feasibility of the adjustment strategy, Luan et al. (2018a) proposed a model that jointly optimizes timetable management and speed control. The train dynamic was constructed by taking a block in a section as a unit, with speed and arrival (depart) decision variables. The linear approximation method was adopted for the constraint containing the quadratic terms to reduce the calculation burden. This method can gain a rescheduled timetable and speed trajectories of each train in sections. On this basis, energy consumption is also considered in Luan et al. (2018b). The adjustment result is a compromise between saving energy and reducing delays. Li et al. (2019) proposed a model in case of a temporary speed restriction. The speed restriction is caused by bad weather, which is modeled as if-then. It caused the rescheduling problem to become nonlinear programming (NLP). The NLP model is converted by linearization to MILP and then solved by CPLEX. Based on the quasi-moving block rule, Xu et al. (2017a) utilized the AG theory of conducting a train rescheduling model in case of temporary speed restriction from a microscopic perspective. This model can adjust the number of blocks between the two trains according to the speed of the two trains. The two-stage solution is utilized to speed up the calculation. That is, the solution is performed in two steps. The solution of the model only considers a part of the constraints taken as the initial value of the second stage. The model in the second stage considers all the constraints. The final result is obtained by solving the entire model with the initial solution Kersbergen et al. (2016) devised a distributed traffic management method. This method splits a railway

network into several regions according to the optimal partition calculation method. According to whether rescheduling globally or locally and border weights, four distributed MPC are proposed to solve the train rescheduling problem. Compared with centralized MPC, which solves the problem of the whole network, distributed MPC is more efficient. Besides train timetables, passengers are also considered in D'Ariano et al. (2017). The macroscopic passenger assignment model is integrated into the microscopic train rescheduling model. The Game Theory-based method can find a trade-off between minimizing train delays and passenger traveling time. Pellegrini et al. (2014) proposed a fine-grained model that considers the infrastructures for complex junctions. By comparing the results of models with different granular, they found that granularity greatly influences the solution results. Fine granularity shows better solution quality than a rough one. Jiang et al. (2017) proposed a train timetabling method in highly congested lines with skip-stop planning. The times caused by accelerating and decelerating are considered in changing the stop plan. Sama et al. (2017) proposed a variable neighborhood search method that decomposes the real-time traffic control problem into train rescheduling and rerouting. This meta-heuristic algorithm shows relatively high solving efficiency compared with a commercial MILP solver and tabu search. Krasemann (2012) proposed a depth-first search to solve the train rescheduling problem. This greedy algorithm can search the shortest running time in stations and be implemented in parallel, which speeds up the search.

Zhan et al. (2015) proposed a MILP model for complete blockages, and they added valid inequalities about the logical relationship between 0–1 variables to reduce solution space. The two-stage solution was adopted, in which the constraints on station capacity were removed in the first stage. Cavone et al. (2020) integrated distributed optimization and bi-level heuristic method in MPC to deal with large-scale railway networks in case of disruption and disruption. Zhang et al. (2020a) conducted a model that collaboratively optimizes train rescheduling and tracks emergency maintenance. The nonlinear constraints about the type of maintenance were linearized. The Lagrangian relaxation algorithm was adopted to reduce the burden of calculation. The sub-problems gained by decomposition can be computed in parallel. Zhu and Goverde (2019) added a flexible short-turning strategy in the MILP model in case of disruption. This model involved changing running time about accelerations(decelerations) due to additional dwell. Besides considering the timetable's periodicity, the rolling stock circulation at terminal stations was also considered. Zhu and Goverde (2021) proposed sequent and combined models to cope with multiple connected disruptions. The rolling horizon optimization was adopted to deal with the long-duration disruption problem. The results showed that the combined model was better than the sequent model. Ata et al. (2019) used the resource conflict graph to construct an ILP model. They split the railway line into several regions to accomplish coordinated rescheduling in multi-regions. The Lagrangian relaxation algorithm was used after solving all regions individually to eliminate the conflicts in boundaries. In train rescheduling for disruption, the passenger is also a key factor. Zhan et al. (2021) proposed a model that optimizes timetable and passenger routes simultaneously. The problem was modeled as an ILP and based on the space–time network. Bi-level dual decomposition was utilized in the solution, which obtained a satisfying result after relatively few iterations.

2.2. Train timetabling and platforming

Trains travel through stations according to the platform assignment decision. The platform assignment guides which routes and platform a train enters and exits when they visit a station. Each train is assigned an inbound and outbound route and platform in a station. Platforming is similar to the train timetable rescheduling problem. It solves the conflicts between trains in the station schedule to ensure safety. The platforming problem has been implicitly considered with the decision about routing(rerouting). Meng and Zhou (2014b) proposed a model based on a time–space network and cumulative flow variables, which accomplish rerouting and rescheduling simultaneously on an N-track network. Lagrangian relaxation-based solution decomposes the constraints about cell capacity, which can gain a lower bound to generate feasible solutions for the proposed model. In Meng and Zhou (2019), they considered integrating the passenger demand, timetable, and routing to increase the profit of the train service. Furthermore, the problem is solved by Lagrangian Relaxation (LR). Binder et al. (2017) proposed a multi-objective rescheduling model with the rerouting strategy. The objective functions included passengers, the cost of operation, and delay. By utilizing the $\sigma-$ constraints, the problem was converted into an ILP, and the Pareto frontier was explored. Pellegrini et al. (2014) constructed a MILP model for train routing and scheduling in complex junctions. In the detail of the station, the maximum headway between a potential conflict can be calculated based on the fix-speed model. In Pellegrini et al. (2019), they constructed valid inequalities related to the routing and scheduling decision variables.

Cacchiani et al. (2015) combined the train timetabling problem and platforming problem for the non-periodic train. The proposed model was solved in sequence due to its complexity. Lu et al. (2022) proposed a train platforming and rescheduling model. Based on the concept of "Degree of Conflict (DoC)", the model can be used in various interlocking mechanisms with an aggregate approach. Mannino and Andreas (2023) proposed to extend the job-shop scheduling problem to train scheduling with platforming for large passenger stations. They also proposed an iteration-based solution approach to overcome the large-scale model's computation burden. The Oslo Central Station test instances demonstrate that the proposed method can solve the problem in a reasonable time. Considering the station section maintenance, Zhang et al. (2019a) proposed an integration model of train timetabling and track maintenance scheduling in microscopic. To overcome the burden of computation, they utilized an iteration framework. The original problem was decomposed as 'train scheduling and track maintenance scheduling' and 'train scheduling and rerouting.' The first sub-problem outputs maintenance task start and end times to the second sub-problem. Furthermore, the latter gives the fixed train routes to the first. The approximate optimal solution can be obtained by iterating between the two. Zhang et al. (2020b) proposed a re-optimization approach for train timetable and platform schedule in case of planned track maintenance. The space–time network conducts an integer programming model for train rescheduling with station routing. The LR and ADMM dualize the constraints about headway and track capacity. Moreover, the result shows that the proposed model can use the infrastructure more efficiently,

Table 1
The difference between publications on train rescheduling with uncertainty.

Details	Uncertainty	Problem(s)	Method	Solution
Nielsen et al. (2012)	Duration of the disruption	Timetable Rolling stock	Rolling horizon Optimization	cplex
Zhan et al. (2016)	Duration of the disruption	Timetable	Rolling horizon Optimization	cplex
Li et al. (2016)	Passenger arrival	Timetable Stop plan	State-constrained control method	MPC
Zhu and Goverde (2020)	Duration of the disruption	Timetable Rolling stock	Two-stage stochastic optimization	cplex
Cacchiani et al. (2020)	Passenger demand	Timetable Rolling stock	Additional robustness in objective function	cplex
Zhang et al. (2021b)	Speed restriction	Timetable	Scenario-based chance constraints	MPC
This paper	Speed restriction	Timetable Platform assignment	Scenario-based chance constraints	MPC & ADMM

and the ADMM method performs better, especially in large-scale problems. In Zhang et al. (2022b), they extend the model to timetable and platform rescheduling with maintenance scheduling decisions. Xu et al. (2017b) considered platform choice with dynamic passenger flow demand in train timetabling based on the time–space network to improve the convenience of passengers and operation efficiency.

Train timetable rescheduling with considering platforming is still challenging due to the complex topology of the stations and numerous potential conflicts. Though part of the papers focused the train timetable rescheduling by considering platforming at a specific station, it is challenging to get the approximate global optimal adjustment result for an entire HSR line. The stations play a decisive role in the line operation efficiency. Efficiently utilizing the stations' infrastructure is essential to reduce train delays and restore regular train operation.

2.3. Traffic management with uncertainty

Nielsen et al. (2012) utilized the rolling horizon-based heuristic framework to solve the large-scale train rescheduling problem. The uncertain duration can be solved by adjusting the horizon according to the updated information about the disruption. In Peng et al. (2016), Zhan expanded the former model in Zhan et al. (2015). The events and activities of trains in different directions were considered. To deal with the uncertain duration of disruptions and reduce the scale of the problem, he utilized rolling horizon optimization to decompose the problem and adjust the horizon dynamically. Zhang et al. (2021b) solved the uncertain impact of the speed restriction by constructing scenario-based chance constraints. By linearization, the chance constraints were converted to deterministic constraints. They also proposed scenarios-based chance-constrains MPC to realize closed-loop adjusting. The proposed model and method can decrease the total delay and headway compared to other rescheduling strategies, such as FCFS and FSFS, and deterministic scenarios. To dynamically and robustly reschedule the timetable in case of uncertain duration disruption, Zhu and Goverde (2020) proposed a two-stage stochastic model to compute the results of all scenarios. The stochastic method can generate a better rescheduling solution with fewer delays and cancellations than the deterministic method. Cacchiani et al. (2020) considered the additional passenger uncertain demand and proposed three optimization models considering robustness. These models can jointly optimize the train stop plan and timetable according to the distribution of passenger demand. Li et al. (2016) constructed a state-constrained control model for the circle line metro to deal with the uncertain arrival passenger flow. This model can minimize the operation cost and decrease the impact of uncertainty. Cadarso and Marín (2014) added terms about punctuality and stock matching in the model to improve the robustness of rolling and connecting rolling stock in a rapid transit network. Making minor delays can promote the robustness of rolling stock circulations. Su et al. (2017) constructed a multi-layer optimization model for infrastructure maintenance planning. The top-level considered the chance constraints of the infrastructure maintenance and used MPC to control it. The threshold between maintenance and fixed costs can be obtained in the middle level. Each maintenance plan can be optimized according to the results of the upper two layers. Table 1 summarizes the related traffic management research with uncertainty.

3. Problem description

This paper considers solving the joint rescheduling problem for train timetabling and platforming. Each direction of the railways is taken into account. In sections, trains cannot utilize the reverse tracks. In stations, trains are allowed to use the reverse routes and platforms. All stations are composed of switches, routes, and platforms. For example, in Fig. 2, the layout of station s and $s+1$ are displayed.

The switches decide the set of available routes and platforms of trains. Besides, the running times on the routes are different due to the routes' length and the platforms' position. Here, the running time on a route (inbound or outbound) is fixed. It is not related to the trains running on it. As shown in Fig. 3, two routes are included. Each train goes through four processes in the station: arriving at the inbound route, arriving at the platform (regarded as departing from the inbound route), departing from the platform (regarded as arriving at the outbound route), and departing from the outbound route respectively. The time that the train arrives at

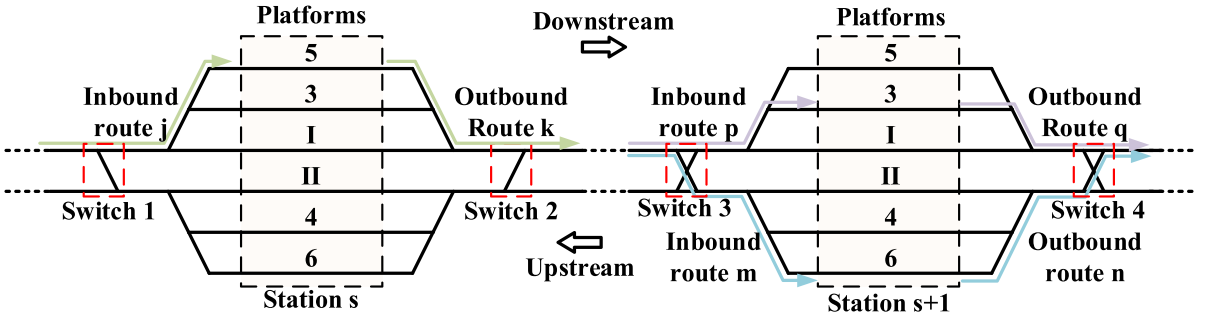


Fig. 2. Schematic diagram of stations structure.

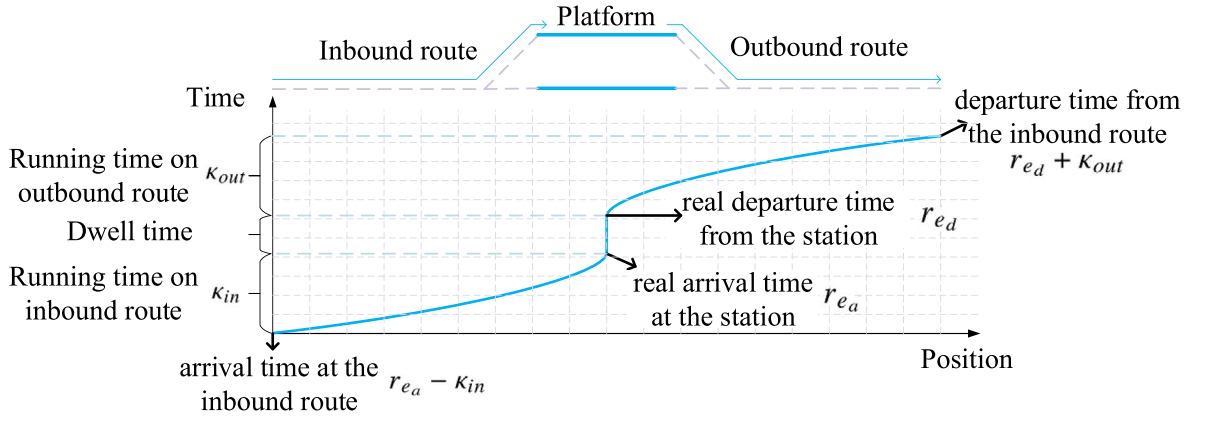


Fig. 3. The fixed running time on a route in stations.

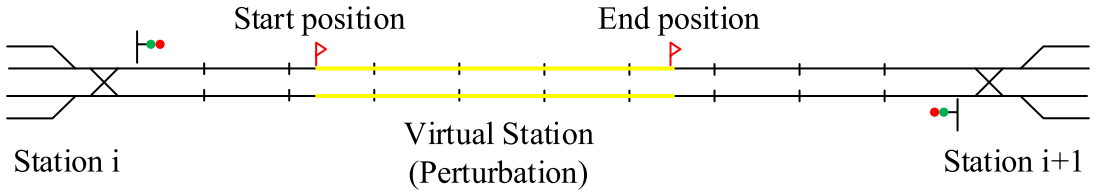


Fig. 4. The virtual station caused by perturbations.

and departs from the platform is the real arrival time at the station and the real departure time from the station, which are denoted by the symbols r_{e_a} and r_{e_d}. The blue curves at the bottom of Fig. 3 represent the process of the train running inside the station. The first and last parts are the process of running on the inbound and outbound routes. The second part, the vertical line, denotes the train dwelling at the track. Hence, the time the train arrives at the inbound routes and departs from the outbound route can be calculated with the fixed running time on routes κ_{in} and κ_{out}. The times that a train arrives at the inbound route and departs from the outbound route are $r_{e_a} - \kappa_{in}$ and $r_{e_d} + \kappa_{out}$. In the actual HSRs, there are different kinds of trains on a line; the main difference is in speed. For example, trains with a speed of 350 km/h and 300 km/h run on China's high-speed rail lines simultaneously. Trains always run at a relatively consistent speed inside the station because of the static speed restriction. Therefore, a train's running time on a route only relates to its chosen route.

To be more precise, the perturbations caused by strong winds lead to temporary speed restrictions, resulting in trains having to decelerate. The effect of perturbations is modeled as the additional running time in the affected sections. This information is displayed on the disaster warning system, which uses field-placed sensors in the train dispatching system to identify the affected sections. The section is considered a virtual station, as shown in Fig. 4. Unlike regular stations, virtual stations have only one platform in each direction and no switches or routes. Therefore, trains cannot overtake each other in virtual stations, nor can they occupy the reverse platforms. The scheduled running time on the affected section is the scheduled dwell time in this virtual station. Additionally, the additional running time caused by perturbation is regarded as the additional dwell times in the virtual station.

To study the joint optimization for timetabling and platforming of HSRs more rigorously, several assumptions are made:

1. The running time on a route is constant.

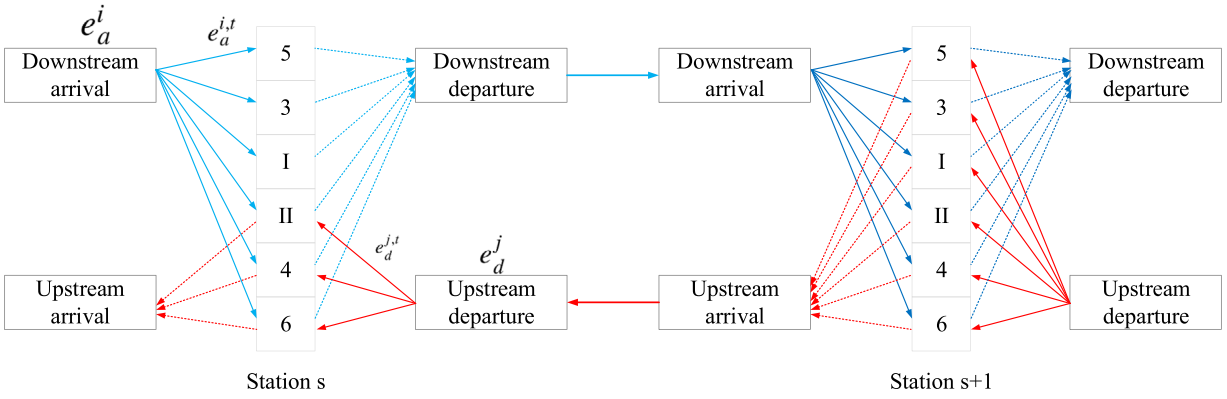


Fig. 5. The division of scheduled event.

2. A track is linked to only one inbound and one outbound route.
3. The train has to dwell after rescheduling if it has a scheduled dwell plan.

4. Mathematical formulations

This section proposes a joint rescheduling model for train timetabling and platforming based on an event-activity network. The first introduced event-activity network serves as the model's foundation. Then the definition of parameters and decision variables are listed. Meanwhile, the objective function and constraints are constructed. To address the uncertainty of perturbations, we incorporate chance constraints into our model and discuss how to generate scenarios in uncertain situations in the last two parts.

4.1. Event-activity network

This paper uses the event-activity network to describe and model the joint rescheduling problem. The event-activity network is a directed graph $G = (E, A)$. E denotes the set of points, and A denotes the set of edges. In train rescheduling, a point represents an event (arrival/departure), and an edge represents an activity between two different events.

1. Event
   An event can represent either a train's arrival or departure event. In the planned timetable, a train's arrival event, departure event, and platform in a station have been scheduled. We allow trains to change the routes and platforms. Therefore, in the rescheduling model, a scheduled event is divided into several possible events according to the available routes and platforms. Taking stations in Fig. 2 as an example, the scheduled events in stations are different due to the difference in layout. As shown in Fig. 5, the two stations' scheduled downstream arrival and departure events are divided into six possible events. For example, the scheduled arrival event e_a^i is divided into $R_{e_a^i} = \left\{e_a^{i,t}, t \in \{I, II, 3, 4, 5, 6\}\right\}$. $R_{e_a^i}$ is regarded as the set of possible events of the scheduled event e_a^i. The scheduled events can take place on any selectable platform after rescheduling. The blue solid and dashed lines represent the possible arrival and departure events. In upstream, the arrival and departure events can also be divided into six possible events in station $s+1$. However, in station s, the scheduled arrival and departure events can only be divided into three possible events. The station's layout s makes the arrival and departure events only occur on platform II, 4, and 6.
2. Activity
   An activity is composed of two different events and connects them. There are five types of activity.

   (a) Train activity in station $q = (e_a, e_d) \in A_{sta}$: this activity is made up of a scheduled arrival e_a and a scheduled departure event e_d of a train in stations. The q can be a dwell activity(the departure event takes place later than the arrival event) or a pass-through activity(arrival and departure events coincide).
   (b) Train activity in section $q = (e_d, e_a) \in A_{sec}$: a train activity in the section comprises a scheduled departure event e_d at station i and a scheduled arrival event e_a at the following station j of a train.
   (c) Headway activity in station $q = (e_a^i, e_a^j) \in A_{arr}^H$ ($q = (e_d^i, e_d^j) \in A_{dep}^H$): this activity involves two scheduled arrival (e_a^i and e_a^j) (departure (e_d^i and e_d^j)) events in the same station of two trains (train i and j) in the same direction. For example, the symbol q in Fig. 7 is a headway activity in the station.
   (d) Headway activity of reverse running in station $q = (e^i, e^j) \in A_{rr}^H$:this kind of activity is defined as two trains in different directions. It is composed of a scheduled arrival event, and a scheduled departure event of them, and their possible events are in conflict. For example, in Fig. 5, because the possible event of e_a^i conflicts with the possible event of e_d^j, such as $e_a^{i,II}$ and $e_d^{j,II}$. The scheduled event e_a^i and e_d^j can form a headway activity of reverse running in a station. The symbol q' in Fig. 7 is a headway activity of reverse running in a station.

Table 2
The definition of parameters.

Symbol	Definition
S, S', T	The set of stations, sections and trains
T^U, T^D	The set of trains in upstream and downstream
E	The set of scheduled events
$E^{sta}_{arr}, E^{sta}_{dep}$	The set of scheduled arrival (arr) and departure (dep) events at station sta
A	The set of activities
A_{sta}	The set of train activities in station sta
A_{sec}	The set of train activities in section sec
A^H_{arr}, A^H_{dep}	The set of arrival and departure headway activities
A^H_{rr}	The set of headway activity of reverse running in station
A^H_{plat}	The set of headway activity on platform
R_e	The set of possible events of the scheduled event e, the set of available platform of e
Γ	The set of perturbations
s_v	The index of perturbation and virtual station
P_e	The scheduled time of event e
ω_q	The minimum duration of the activity q
ω_{s_v}	The additional running time due to the perturbation s_v
$\kappa^t_{in}, \kappa^t_{out}$	The running time on the inbound and outbound routes that link to platform t
$\overline{\omega}, \overline{\omega}$	The start and end time of the disturbance
M	A sizable positive integer (larger than 1440)

Table 3
The definition of decision variables.

Symbol	Definition	Type
r_{e^t}	The time that possible event e^t occurs $\forall e^t \in R_e, e \in E$	Integer
λ_{e^t}	Whether the possible event e^t occurs $\forall e^t \in R_e, e \in E$	Binary
τ_q	Whether q is a dwell activity $\forall q \in A_{sta}$	Binary
γ_q	Whether the activity q is affected by perturbation at virtual station S_v $\forall q \in A_{sta}, sta = Sv$	Binary
ζ_q	The occurrence order of the two events that make up q $\forall q \in A^H_{arr} \cup A^H_{dep} \cup A^H_{rr} \cup A^H_{plat}$	Binary

(e) Headway activity on platform $q = (e^{i,t}, e^{j,t}) \in A^H_{plat}$: this activity connects two possible events that are in conflict, $e^{i,t}$ and $e^{j,t}$. The symbol q'' in Fig. 7 is a headway activity on the platform.

4.2. The joint rescheduling model

The main goal of the proposed model is to decrease delays and resume the regular operation plan as soon as possible after perturbations occur. The objective function is the total delay of all scheduled arrival and departure events. The definition of parameters and decision variables are shown in Tables 2 and 3.

Objective function

$$\min J = \sum_{e \in E} \Big(\sum_{e^t \in R_e} r_{e^t} - P_e \Big) \tag{1}$$

Constraints of track selection in a station

1. As described in Section 4.1, each scheduled event e is divided into several possible events e^t ($e^t \in R_e$, R_e is the set of all possible events of event e) in the rescheduling model according to the set of available platforms. Only one of them can take place. Constraints (2) guarantee the above.

$$\sum_{e^t \in R_e} \lambda_{e^t} = 1 \quad \forall e \in E \tag{2}$$

If event e^t occurs, the scheduled event e will occur on the platform t after rescheduling. Then $\lambda_{e^t} = 1$ and r_{e^t} will be greater than 1. For other possible events, $r_{e^t} = \lambda_{e^t} = 0$. These constraints are satisfied by constraints (3).

$$\lambda_{e^t} \le r_{e^t} + (1 - \lambda_{e^t}) \times M \le M \quad \forall e \in E, \forall e^t \in R_e \tag{3}$$

A train's arrival event e_a and departure event e_d must occur on the same track, as shown in constraints (4).

$$\lambda_{e^t_a} = \lambda_{e^t_d} \quad \forall sta \in S, \forall (e_a, e_d) \in A_{sta}, \forall e^t_a \in R_{e_a}, \forall e^t_d \in R_{e_d}, \tag{4}$$

2. In a station, an activity $q \in A_{sta}$ can be a dwell activity or passing through activity. This model involves additional dwell. However, skip-stopping is not allowed to guarantee that passengers can get on and off. $\forall q = (e_a, e_d) \in A_{sta}$, if it is originally

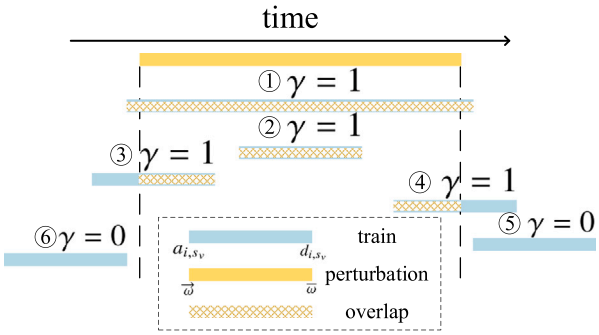


Fig. 6. Judgement on if a train is affected.

a dwell activity, which means that $P_{e_a} < P_{e_d}$, it will still be a dwell activity after rescheduling and $\tau_q = 1$. If $P_{e_a} = P_{e_d}$, τ_q can be equal to 0 or 1. Constraints (5) ensure the above requirement.

$$0 \le P_{e_d} - P_{e_a} + (1 - \tau_q) \times M \le M \quad \forall sta \in S, \forall q = (e_a, e_d) \in A_{sta} \tag{5}$$

3. The minimum dwell time should be promised if the activity q is a dwell activity. Constraints (6) make this happen.

$$\begin{aligned} &\sum_{e_d^t \in R_{e_d}} r_{e_d^t} - \sum_{e_a^t \in R_{e_a}} r_{e_a^t} \ge 0 \\ \omega_q \le &\sum_{e_d^t \in R_{e_d}} r_{e_d^t} - \sum_{e_a^t \in R_{e_a}} r_{e_a^t} + (1 - \tau_q) \times M \\ &\forall sta \in S, \forall q = (e_a, e_d) \in A_{sta} \end{aligned} \tag{6}$$

4. In the real world, the sidelines of stations are generally used for train stopping and passengers getting on and off. Trains always pass through the stations from the mainlines. Thus, the events related to dwelling activities must occur on the sidelines. A train must select the mainline if it passes through a station. Thus, one of train occupying the mainline and train dwelling can occur. In constraints (7), R_e^m is the set of possible events that can take place on the mainlines of event e, the above is modeled as the following.

$$\tau_q + \sum_{e_a^t \in R_{e_a}^m} \lambda_{e_a^t} = 1 \quad \forall sta \in S, \forall q = (e_a, e_d) \in A_{sta} \tag{7}$$

Constraints of the scheduled timetable

In order to ensure passengers have enough time to get on and off, the rescheduled time must be greater than the scheduled time.

$$\sum_{e^t \in R_e} r_{e^t} \ge P_e \quad \forall e \in E \tag{8}$$

The impact of perturbations on operation

1. The section where perturbation occurs is regarded as a virtual station. In Fig. 6, a_{i,s_v} and d_{i,s_v} are the arrival and departure times of train i at this virtual station. Here, a binary decision variable γ is proposed to indicate whether a train is affected by perturbation at the virtual station s_v. It can be divided into 6 scenarios according to $[\overline{\omega}, \overline{\omega}]$, which are the start and end times of the perturbation, respectively, as shown in Fig. 6. The blue rectangle is the rescheduled time at the virtual station, and the yellow rectangle represents the time range of the perturbation. Here, two binary decision variables are introduced. $\gamma_{e_a}^1$ and $\gamma_{e_a}^2$ decide whether train arrives at virtual station later than $\overline{\omega}$ and depart earlier than $\overline{\omega}$ respectively. Constraints (9) and (10) express the above explanation.

$$0 \le \overline{\omega} - r_e + (1 - \gamma_{e_a}^1) \times M \le M \quad \forall s_v \in \Gamma, \forall e_a \in E_{arr}^{s_v} \tag{9}$$

$$0 \le r_e - \overline{\omega} + (1 - \gamma_{e_a}^2) \times M \le M \quad \forall s_v \in \Gamma, \forall e_a \in E_{dep}^{s_v} \tag{10}$$

2. In scenarios 1–4, the blue rectangles overlap (represented by the brown shade) with the yellow one, which can infer that train i is affected by perturbation and $\gamma_q = 1$. However, in 5–6, the opposite occurs. Thus, once $r_{e_a} \le \overline{\omega}$ or $r_{e_d} \ge \overline{\omega}$, the perturbation does not influence the train. Only when they are both equal to 1, γ_q can be equal to 1, which is guaranteed by constraints (11).

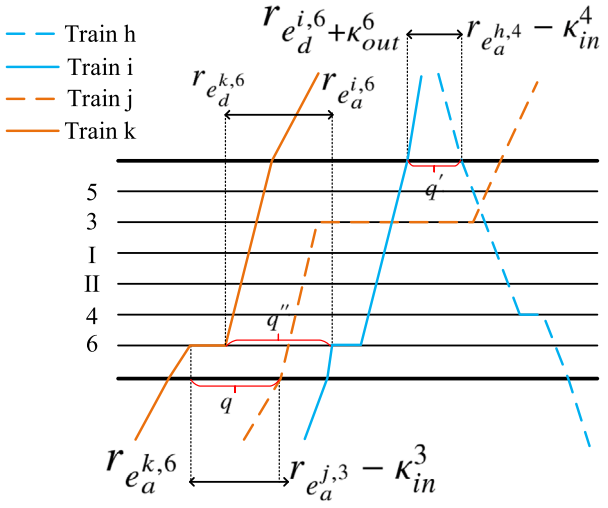


Fig. 7. The headway between two trains.

$$\begin{aligned} \gamma^1_{e_a} &\geq \gamma_q \\ \gamma^2_{e_d} &\geq \gamma_q \\ \frac{\gamma^1_{e_a} + \gamma^2_{e_d}}{2} - 0.5 &\leq \gamma_q \\ \forall s_v \in \Gamma, \forall e_a \in E^{s_v}_{arr}, \forall e_d \in E^{s_v}_{dep}, q = (e_a, e_d) &\in A_{s_v} \end{aligned} \tag{11}$$

Constraints of minimum running time in sections

These constraints guarantee that the running time in sections will not be violated.

1. The running time on the section cannot be less than the minimum time to ensure safe operation. Constraints (12) guarantee the above condition.

$$\sum_{e^t_a \in R_{e_a}} (r_{e^t_a} - \lambda_{e^t_a} \times \kappa^t_{in}) - \sum_{e^t_d \in R_{e_d}} (r_{e^t_d} + \lambda_{e^t_d} \times \kappa^t_{out}) \geq \omega_q \quad \forall sec \in S', \forall q = (e_a, e_d) \in A_{sec} \tag{12}$$

   Here, for a section activity, the symbol ω_q represents the minimum running time from the end of the outbound route to the start of the inbound route at the next station.
2. According to constraints (9)–(11), a binary variable indicates whether the train is affected by speed restriction can be gained. If a train is affected, it should dwell longer at the virtual station. The minimum dwell time is the additional running time caused by speed restriction plus the minimum running time on the speed restriction section. It is guaranteed by constraints (13). The virtual station is composed of two platforms. Divining the scheduled arrival and departure events in the virtual station is unnecessary.

$$r_{e_a} - r_{e_d} \geq \gamma_q \omega_{s_v} + \omega_q \quad \forall s_v \in \Gamma, \forall q = (e_a, e_d) \in A_{s_v} \tag{13}$$

Constraints of the headway between trains

1. There should be a safe headway between arrival and departure events for trains in the same direction. As shown in Fig. 7, the solid orange line and the dotted orange line, respectively, refer to the running processes of train k and train j in the station. The station here has the same structure as the station $s+1$ in Fig. 2. The thick lines represent the two ends of the station. For an upstream train, the thick line at the bottom represents the start of the inbound route, and the thick line at the top denotes the end of the outbound route. The opposite is valid for a downstream train. In this figure, train k and j occupy tracks 6 and 3, respectively. They have conflicts with arrival and departure. The arrival times at the station of train k and j are denoted by symbols $r_{e^{k,6}_a}$ and $r_{e^{j,3}_a}$. For example, a conflict between train j and k can occur at their inbound routes m and p in Fig. 2. Hence, train k can only arrive at the inbound route m after the inbound route p is released for a while. That is $r_{e^{j,3}_a} - \kappa^3_{in} - r_{e^{k,6}_a}$ must be larger than minimum headway. When considering arrival or departure order in this type of activity, the constraints can be constructed as Eq. (14). Where, $\overline{\kappa^{i,t}} = \lambda_{e_{i,t}} \times \kappa^t_{out} \times (1 - \zeta_q)$ if the activity $q \in A^H_{dep}$. $\overline{\kappa^{j,t}} = \lambda_{e_{j,t}} \times \kappa^t_{out} \times \zeta_q$. Otherwise, $\overline{\kappa^{i,t}} = -\lambda_{e_{j,t}} \times \kappa^t_{in} \times \zeta_q$, and $\overline{\kappa^{j,t}} = -\lambda_{e_{j,t}} \times \kappa^t_{in} \times (1 - \zeta_q)$. The symbol $\overline{\kappa^{j,t}}$ is non-linear. The linearization and analysis

processes of constraints (14) are shown in Appendix A.

$$\begin{aligned}\omega_q + \sum_{e^{j,t}\in R_{e^j}} (r_{e^{j,t}} + \overline{\kappa^{j,t}}) \le \sum_{e^{i,t}\in R_{e^i}} (r_{e^{i,t}} + \overline{\kappa^{i,t}}) + (1-\zeta_q)\times M \\ \omega_q + \sum_{e^{i,t}\in R_{e^i}} (r_{e^{i,t}} + \overline{\kappa^{i,t}}) \le \sum_{e^{j,t}\in R_{e^j}} (r_{e^{j,t}} + \overline{\kappa^{j,t}}) + \zeta_q \times M \\ \forall q = (e^i, e^j) \in A^H_{arr} \cup A^H_{dep}\end{aligned} \tag{14}$$

2. In sections, trains cannot overtake, guaranteed by constraints (15). Y_s denotes a set of activity pairs, $\forall \upsilon \in Y_s$, υ connects activities q_1 and q_2. q_1 and q_2 represent the departure headway activity of two trains and their arrival headway activity at the next station, respectively.

$$\zeta_{q_1} - \zeta_{q_2} = 0 \quad \forall \upsilon \in Y_s \tag{15}$$

3. If two trains select the same platform, the headway between the arrival time at the platform of the following train and the departure time from the platform of the leading train should be greater than the minimum one. ζ_q is meaningful only both $\lambda_{e^{i,t}}$ and $\lambda_{e^{j,t}}$ are equal to 1. In other words, only when both trains occupy the same platform does it make sense to consider the minimum headway between them. Otherwise, the constraints (16) are non-active.

$$\begin{aligned}\omega_q + r_{e^{j,t}} \le r_{e^{i,t}} + (3 - \lambda_{e^{i,t}} - \lambda_{e^{j,t}} - \zeta_q)\times M \\ \omega_q + r_{e^{i,t}} \le r_{e^{j,t}} + (2 - \lambda_{e^{i,t}} - \lambda_{e^{j,t}} + \zeta_q)\times M \\ \forall q = (e^{i,t}, e^{j,t}) \in A^H_{plat}\end{aligned} \tag{16}$$

Besides, overtaking cannot happen on the tracks.

$$\zeta_{q_1} = \zeta_{q_2} \quad \forall q_1 = (e^{i,t}, e^{j,t}) \in A^H_{plat}, q_2 = (e^{i',t}, e^{j',t}) \in A^H_{plat} \tag{17}$$

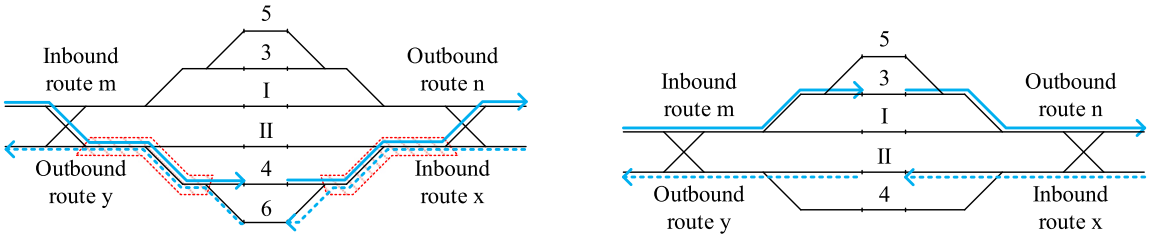
The symbol i and i' refer to the same train. If $e^{i,t}$ is an arrival event, then $e^{i',t}$ is a departure event. The set of activity pairs, Y_r, is defined, $\forall \upsilon \in Y_r$, $\upsilon = (q_1, q_2)$. For a special subset Y^o_r of Y_r, $\forall \upsilon \in Y^o_r$, $\upsilon = (q_1, q_2)$, the trains that are relate to υ are in different directions. To be noticed is that $\forall q = (e^{i,t}, e^{j,t}) \in A^H_{plat}$, ζ_q considered to be taken as 1 or 0 only when $\lambda_{e^{i,t}}$ and $\lambda_{e^{j,t}}$ are both equal to 1. In other words, if the two trains do not select the same platform, considering the relationship between their arrival and departure events on the platform is meaningless, the constraints (18) can be obtained.

$$\begin{aligned}\lambda_{e^{i,t}} \ge \zeta^q \\ \lambda_{e^{j,t}} \ge \zeta^q \\ \forall q = (e^{i,t}, e^{j,t}) \in A^H_{plat}\end{aligned} \tag{18}$$

$\sum^R \zeta_q$ denotes the order between the arrival and departure event of two trains, and R is the set of tracks that are available for both. If a train, i, arrives at the station after another train, j, departs from this station, it can be inferred that the train i arrives and departs later than the train j. Hence, the constraints (19) can be obtained.

$$\zeta_{q_1} \ge \sum^R \zeta_{q_2} \quad \forall q_1 = (e^i, e^j) \in A^H_{arr} \cup A^H_{dep}, q_2 = (e^{i,t}, e^{j,t}) \in A^H_{plat} \tag{19}$$

4. The headway activity of reverse running in the station should be considered specially. In Fig. 8(a), the solid blue lines denote the routes selected by the downstream train. Moreover, the blue dotted lines represent the routes an upstream train selects. The upstream train selects the reverse routes and platform. Therefore, conflicts can happen at the red-shaded part. No trains select the reverse routes and platform in Fig. 8(b). They do not interfere with each other. The train h and i in Fig. 7 can represent the scenario in Fig. 8(a). The train i departs earlier than train h arrives. Furthermore, train h selects the reverse routes and platform. Due to the existence of the interlocking mechanism, only after the outbound route y selected by train i is released train h can only occupy the inbound route m. Thus, the symbol $r_{e^{h,4}_a} - \kappa^4_{in} - (r_{e^{i,6}_d} + \kappa^6_{out})$ should be larger than the minimum headway. In order not to lose generality, if the order between trains is considered, the constraints can be constructed as Eqs. (20) and (21). In constraints (20), the symbol $\overline{\kappa^{i,t}}$ is non-linear. It is equally to $-\lambda_{e^{i,t}} \times \kappa^t_{in} \times \zeta_{q_1}$ if the event e_i is an arrival event. Otherwise, it is equal to $\lambda_{e^{i,t}} \times \kappa^t_{out} \times \zeta_{q_1}$. The linearization and analysis of constraints (20) are similar to the constraints (14) in Appendix A. The symbol e represents the departure-related event of both e^i and e^j in constraints (20), R^r_e denotes the set of possible events of e that take place on the reverse platforms. If $\sum_{e\in R^r_e} \lambda_e = 1$, the train related to event e selects the reverse routes and platform. If $\sum_{e\in R^r_e} \lambda_e = 0$, the train does not select the reverse platform, and the constraints (20) and (21) is invalid. The symbol $A^H_{\overleftarrow{rr}}$ is a subset of A^H_{rr}, $\forall p = (e^i, e^j) \in A^H_{\overleftarrow{rr}}$, $e^i \in E^{sta}_{arr}$ and $e^j \in E^{sta}_{dep}$. $\forall p = (e^i, e^j) \in A^H_{\overleftarrow{rr}}$,



(a) Case 1 (b) Case 2

Fig. 8. The potential conflicts between trains in different directions.

$e^i \in E^{sta}_{dep}$ and $e^j \in E^{sta}_{arr}$.

$$\begin{aligned}
&\omega_{q_1} \times \sum_{t \in R^r_e} \lambda_{e^t} \le \sum_{e^{i,t} \in R_{e^i}} (r_{e^{i,t}} + \overrightarrow{\kappa^{i,t}}) - \sum_{e^{j,t} \in R_{e^j}} (r_{e^{j,t}} + \overrightarrow{\kappa^{j,t}}) + (1 - \zeta^{q_1}) \times M \\
&\omega_{q_2} \times \sum_{t \in R^r_e} \lambda_{e^t} \le \sum_{e^{j,t} \in R_{e^j}} (r_{e^{j,t}} + \overrightarrow{\kappa^{j,t}}) - \sum_{e^{i,t} \in R_{e^i}} (r_{e^{i,t}} + \overrightarrow{\kappa^{i,t}}) + (1 - \zeta^{q_2}) \times M \\
&\qquad \forall q_1 = (e^i, e^j) \in A^H_{\overrightarrow{rr}}, q_2 = (e^j, e^i) \in A^H_{\overleftarrow{rr}}
\end{aligned} \tag{20}$$

$$\begin{aligned}
&\zeta^{q_1} + \zeta^{q_2} = 1 \\
&\forall q_1 = (e^i, e^j) \in A^H_{\overrightarrow{rr}}, q_2 = (e^j, e^i) \in A^H_{\overleftarrow{rr}}
\end{aligned} \tag{21}$$

Here, a new set of activity pairs Y_{rr} is defined. $\forall \upsilon \in Y_{rr}$, $\upsilon = (q_1, q_2)$.

$$\begin{cases}
\min J \\
s.t. \quad (2)\text{–}(21), \\
\forall e \in E, \quad r_{e^t} \in N, \quad \lambda_{e^t} \in 0, 1 \\
\forall q \in A, \quad \tau_q, \gamma_q, \zeta_q \in 0, 1
\end{cases} \tag{22}$$

4.3. Chance constraints of uncertain perturbations

In the previous research, the speed restriction caused by perturbation is usually considered constant. However, wind speed often changes, which causes the additional running time carried by wind also changes frequently. Scenario-based chance constraints are constructed to cope with this uncertainty and improve the robustness of the model when facing uncertain perturbations. Scenario-based chance constraints are used to describe constraints with strong randomness parameters They improve the robustness of the adjustment result by considering several scenarios, which limits the probability of failure of constraints about random parameters. It allows some variables not to satisfy constraints under certain conditions. However, these variables should guarantee that the probability that they meet the constraints is not less than a certain confidence level α. Eq. (23) is an example. x denotes the decision variable, Φ is the set of all possible scenarios, and ξ is the stochastic parameter vector. For each scenario $\phi \in \Phi$, $g(x, \xi) \le 0$ is a chance constraint. The constraint $P_r(x, \xi) \ge \alpha$ guarantees $g_\phi(x, \xi) \le 0$ is satisfied with at least α.

$$\begin{cases}
\min f(x) \\
s.t. \quad P_r(g_\phi(x, \xi) \le 0, \phi \in \Phi) \ge \alpha
\end{cases} \tag{23}$$

The method of transforming chance constraints into deterministic constraints is utilized. A binary variable μ is introduced here. For a certain scenario ϕ, if $g_\phi(x, \xi) \le 0$, then $\mu = 1$. To be noticed is that $g_i(x, \xi) \le 0$ can also hold when $\mu = 0$. According to this rule, the relationship can be expressed as Eq. (24).

$$g_\phi(x, \xi) \le M(1 - \mu) \tag{24}$$

Here, the deterministic constraints from Eq. (23) can be gained, as shown in Eq. (25).

$$\begin{cases}
\sum_{\phi \in \Phi} P_r(\phi) \mu_\phi \ge \alpha \\
g_\phi(x, \xi) \le M(1 - \mu_\phi) \quad \forall \phi \in \Phi
\end{cases} \tag{25}$$

In this paper, ω_{s_v} is considered the parameter with strong randomness. $\omega_{s_v}^{\phi}$ is the value of ω_{s_v} with the scenario ϕ. Thus, constraints (13) can be rewritten as scenario-based chance-constrained based on the above method:

$$\begin{cases} \sum_{\phi \in \Phi} P_r(\phi)\mu_{\phi}^{q} \geq \alpha \\ r_{e_a} - r_{e_d} - (\omega_q + \gamma_q \omega_{s_v}^{\phi}) \geq M(\mu_{\phi}^{q} - 1) \quad \forall \phi \in \Phi \end{cases} \qquad \forall s_v \in \Gamma, \forall q \in A_{s_v} \tag{26}$$

4.4. Scenarios generation

The probability distribution function can calculate the wind speed. The classical Weibull distribution is used to generate several scenarios and the probability of their occurrence in a perturbation, which has been widely used to calculate the probability of wind speed (see, e.g., Wang and Okaze (2022) and Wang et al. (2022)).

Besides, the wind speed of different perturbations is assumed to be independent and identically distributed. The probability and cumulative distribution functions are shown as Eqs. (27) and (28), respectively. λ is the scale parameter and k is the shape parameter. For a scenario x in the perturbation, the time is discretized with the interval of 0.2 min to calculate its appearance probability. It means that the occurrence probability of x equals the cumulative probability of the wind speed on $[x - 0.1, x + 0.1]$. Hence, $P(x) = F_{x+0.1} - F_{x-0.1}$.

$$f_x = \begin{cases} \frac{k}{\lambda}(\frac{x}{\lambda})^{k-1} e^{-(\frac{x}{\lambda})^k} & x \geq 0 \\ 0 & x < 0 \end{cases} \tag{27}$$

$$F_x = \begin{cases} 1 - e^{-(\frac{x}{\lambda})^k} & x \geq 0 \\ 0 & x < 0 \end{cases} \tag{28}$$

5. Solution algorithm

Train rescheduling is a complex combinational problem, the number of its decision variables and constraints increases quadratically with the number of trains and tracks in stations and linearly with the number of stations, which makes the rescheduling problem become an NP-hard problem. The large scale of the problem dramatically increases the complexity of the computation.

MPC decomposes the large-scale problem into several horizons to ensure a real-time solution. The scale of the entire problem is reduced by solving these sub-problems in each horizon sequentially. This method improves real-time performance. Besides, since trains in different directions are considered for some sub-problems, their scale is still relatively large. ADMM is utilized to dualize part of constraints and solve the slacked problems to construct an upper bound and find the lower bound of the original problems.

The MPC framework is introduced in Section 5.1. The ADMM algorithm is introduced in Section 5.2.

5.1. MPC framework

Model predictive control is based on rolling optimization and feedback correction. It can better solve the optimization control problem with multiple uncertain factors and has strong anti-interference ability and robustness. In this paper, MPC is the basic framework of the solution algorithm. MPC divides the entire time range into several small stages. Each stage is a prediction horizon. Its time range can be denoted as $[h, h + h_p]$, h is the start time, and h_p is the length of the predictive stage. By solving the open-loop control problem in the prediction horizon according to the system's current state, control actions in this stage can be obtained. Moreover, only the actions within the control horizon are executed. The start time of the control horizon is also h. Its time range is $[h, h + h_c]$ and $h_c < h_p$ apparently. MPC does not use a constant global optimization objective but uses a time-forward rolling optimization strategy. It means that optimization is repeatedly completed online.

5.1.1. The rolling process of MPC

The rolling process of MPC is shown in Fig. 9. The two orange rectangles represent two perturbations. Each stage's control and prediction horizons are shown as light and dark blue rectangles. In the real world, the exact time of the end of the perturbation can only be known when the perturbation ends. In this figure, stage 1 is not affected by the first perturbation. However, though stage 2 has an overlapping part with the first perturbation, the perturbation does not occur at the start of stage 2. The rescheduling is triggered when the first perturbation occurs. Stage 3 starts when the perturbation 1 occurs. When constructing the rescheduling model, although from Fig. 9, the perturbation 1 terminates in the period of stage 3, the exact end time of the perturbation 1 is unknowable at the start time of the stage 3. Therefore, the end time of the perturbation 1 is regarded as the end time of the stage 3. So does stage 4. During the control horizon of stage 4, stage 5 starts with the occurrence of perturbation 2. Likewise, the end time of the perturbation 2 is regarded as the end time of the stage 5. In the period of control horizon in stage 5, the perturbation 1 terminates at a specific time. Hence, stage 6 starts at this time. The following rolling procedure of stages 6, 7, and 8 are the same as the above process.

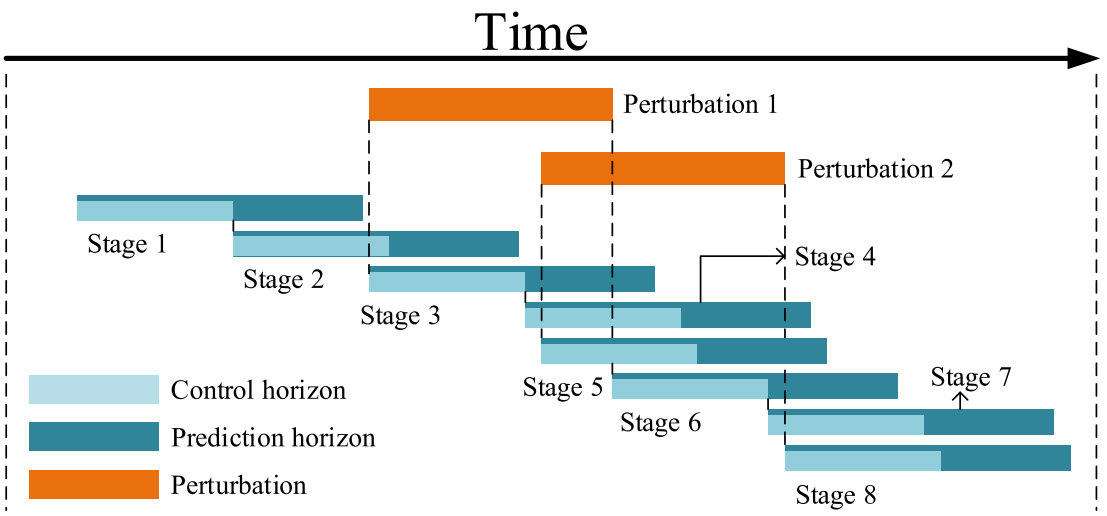


Fig. 9. The rolling process of MPC.

5.1.2. Model construction

The entire optimization is completed by solving the open-loop problem of each prediction horizon iteratively. Hence, the model represented by Eq. (22) has to be constructed and solved several times. However, some constraints in the proposed model are unnecessary for some stages. For example, some trains will be disturbed if a prediction horizon and wind overlap. This stage can be called as disposal stage. The virtual station exists, and trains have to dwell at this station if they are affected by the wind. However, if the wind disappears in a prediction horizon, the wind and the virtual station are redundant in this stage. It is only necessary to resume the planned operation as soon as possible under the premise of safe running without considering the wind. It can be regarded as the recovery stage.

The models in the disposal stage and recovery stage can be constructed as the model (29) and (30), respectively. Here, the constraint set (C_{mpc}) is used to denote the set of all constraints in the model.

$$\begin{cases} \min J \\ s.t. \quad (2)\text{–}(12) \cup (14)\text{–}(21) \cup (26) \in (C_{mpc}) \\ \forall e \in E, \quad r_{e^t} \in N, \quad \lambda_{e^t} \in 0,1 \\ \forall q \in A, \quad \tau_q, \gamma_q, \zeta_q \in 0,1 \end{cases} \tag{29}$$

$$\begin{cases} \min J \\ s.t. \quad (2)\text{–}(8) \cup (12) \cup (14)\text{–}(21) \in (C_{mpc}) \\ \forall e \in E, \quad r_{e^t} \in N, \quad \lambda_{e^t} \in 0,1 \\ \forall q \in A, \quad \tau_q, \gamma_q, \zeta_q \in 0,1 \end{cases} \tag{30}$$

5.2. ADMM algorithm

Trains can occupy the reverse routes and platforms, the constraints about trains in different directions are strongly coupled and numerous, which increase the burden of the computation in each prediction horizon. Here, ADMM is utilized to decompose the large-scale problem. Zhang et al. (2019b) first applied ADMM to solve the cyclic train timetabling problem, which was extended to the integration of line plan and cyclic train timetabling problems in Zhang et al. (2022). Now it has been widely used in train rescheduling problems (see e.g., Zhan et al. (2021), Zhang et al. (2020b) and Luan et al. (2020)). It is a framework for solving optimization problems and uses the Decomposition-Coordination process to decompose an enormous global problem into multiple smaller and more manageable sub-problems. It obtains the solution to the large global problem by coordinating the solutions of the sub-problems. Therefore, the coupled constraints are dualized to decrease the computation burden and generate the proposed model's lower bound. Here, only constraints (17) that relate to trains in different directions and constraints (21) are slacked. The

original model can be rewritten as follow:

$$\begin{cases} \min J = \sum_{e\in E}\left(\sum_{e^t\in R_e} r_{e^t} - P_e\right) \\ + \sum_{v=(q_1,q_2)\in Y_r^o}\left[\theta_v^1(\zeta_{q_1}+\zeta_{q_2}-1)+\frac{\rho_v^1}{2}\left[\zeta_{q_1}+\zeta_{q_2}-1\right]^2\right] \\ + \sum_{v=(q_1,q_2)\in Y_{rr}}\left[\theta_v^2(\zeta_{q_1}-\zeta_{q_2})+\frac{\rho_v^2}{2}\left[\zeta_{q_1}-\zeta_{q_2}\right]^2\right] \\ s.t. \quad (C_{mpc}) \\ \forall e\in E, \quad r_{e^t}\in N, \quad \lambda_{e^t}\in 0,1 \\ \forall q\in A, \quad \tau_q,\gamma_q,\zeta_q\in 0,1 \end{cases} \tag{31}$$

θ^1 is the multipliers of constraints (17) that are related to trains in different directions. θ^2 is the multipliers of the constraints (21). Due to the high symmetry of the upstream and downstream trains, the model (31) can be divided into two models according to directions, which are rescheduling model in upstream ($\aleph_u$) and rescheduling model in downstream ($\aleph_u$). The model ($\aleph_u$) and ($\aleph_d$) are shown as the following.

$$\begin{cases} \min J' = \sum_{e\in E^s}\left(\sum_{e^t\in R_e} r_{e^t} - P_e\right) \\ + \sum_{v=(q_1,q_2)\in Y_r^o}\left[\theta_v^1(\zeta_{q_1}+\zeta_{q_2}-1)+\frac{\rho_v^1}{2}\left[\zeta_{q_1}+\zeta_{q_2}-1\right]^2\right] \\ + \sum_{v=(q_1,q_2)\in Y_{rr}}\left[\theta_v^2(\zeta_{q_1}-\zeta_{q_2})+\frac{\rho_v^2}{2}\left[\zeta_{q_1}-\zeta_{q_2}\right]^2\right] \\ s.t. \quad (C_{mpc}),(C_d) \\ \forall e\in E^s, \quad r_{e^t}\in N, \quad \lambda_{e^t}\in 0,1 \\ \forall q\in A, \quad \tau_q,\gamma_q,\zeta_q\in 0,1 \end{cases} \tag{32}$$

Constraints (C_d) denote the constraints of fixing the decision variables of trains in the opposite direction to the result obtained in the previous iteration. In model ($\aleph_u$), E^s denotes the set of the all scheduled upstream events. It denotes the set of the all scheduled downstream events in model ($\aleph_d$). Due to the existence of $\left[\zeta_{q_1}+\zeta_{q_2}-1\right]^2$ and $\left[\zeta_{q_1}-\zeta_{q_2}\right]^2$, the objective functions are non-linear. Rules of linearization is constructed and introduced in detail in Appendices B and C to decrease the computation burden.

To be noticed is that though constraints (16) are also related to trains in different directions, they are not being dualized because of the existence of M. M causes the ADMM challenging to converge. These constraints still exist in each model. To deal with the strong coupling in these constraints, constraints (16) are taken as an example here. In constraints (16), $e^{i,t}$ and $e^{j,t}$ are thought to be a possible upstream arrival event and a possible downstream departure event, respectively. In upstream model ($\aleph_u$), the downstream decision variables $\lambda_{e^{j,t}}$ and $r_{e^{j,t}}$ are regarded as parameters. All of them are set to be the results of the model ($\aleph_d$) obtained in the last iteration. Hence, the constraints (16) only contain upstream decision variables. So does the downstream model. A detailed ADMM-based joint optimization method is introduced in Algorithm 1.

5.2.1. Dynamic constraints generation

The number of constraints (C_{mpc}) in model ($\aleph_u$) and ($\aleph_d$) is still numerous in some predictive stages. Quite a large part of them are satisfied, and their multipliers are equal to 0, which is unnecessary. Therefore, the dynamic constraints generation method is utilized to deal with the inactive constraints. An empty constraints pool is first constructed in the dynamic constraints generation, and all multipliers are set to zero. After each iteration, all dualized constraints are checked, the violated constraints are put in the constraints pool, and the related multipliers are updated. What needs unique explanation here is that when checking constraints (21), besides judging whether $\zeta_{q_1}+\zeta_{q_2}=1$, whether $\sum_{e^t\in R_e^r}\lambda_{e^t}=1$ also needs to be checked. If $\sum_{e^t\in R_e^r}\lambda_{e^t}=0$, train related to event e does not select the opposing tracks, although $\zeta_{q_1}=\zeta_{q_2}$, there will still be no conflicts. The constraints can only be put into the constraints pool when $\zeta_{q_1}=\zeta_{q_2}$ and $\sum_{e^t\in R_e^r}\lambda_{e^t}=1$. An important issue is emphasized: as shown in Step 3.1 of Algorithm 1, the objectives of the models ($\aleph_u$) and ($\aleph_d$) are reset. Only the constraints in the constraints pool must be considered in their objective functions. This process reduced the complexity of computation and the number of linearizations.

5.2.2. Feasible solution generation

The solutions can be gained by solving the models ($\aleph_u$) and ($\aleph_d$). However, these solutions are only sometimes feasible. These infeasible solutions need to be modified and transformed to be feasible. A new MILP model ($\aleph_f$) is constructed to generate a feasible solution, shown as Eq. (33).

Algorithm 1 ADMM-based joint optimization method.

Input:

The scheduled timetable P;
The information of the perturbation I;
The maximum number of iteration K and gap σ of ADMM.

Output:

The rescheduled timetable r_p;
The rescheduled platform assignment r_s

Step 1. Initialize:

Set the current number of iterations ϱ to 0 and the current gap η to a relatively large positive number, such as M. Construct the model ($\aleph$) according to 5.1.2.

Step 2. Decompose the model and construct the constraints pool:

Decompose the model ($\aleph$) to model ($\aleph_u$) and ($\aleph_d$). Set all multipliers to be 0, the best upper bound $UB^* = +\infty$ and the best lower bound $LB^* = -\infty$. Initialize an empty constraints pool.

Step 3. Main loop:

while $\varrho < K$ and $\eta > \sigma$ **do**

Step 3.1. Generate a solution by ADMM:

Reset the objective function of model ($\aleph_u$) and ($\aleph_d$) and compute them in parallel. Get a solution and the current lower bound LB^{ϱ}.

Step 3.2. Compute the current lower bound:

Use the method in 5.2.2 to transform the solution gained in Step 3.1 into a feasible solution. Get the current upper bound UB^{ϱ}.

Step 3.3. Update the multipliers:

Update the best upper bound $UB^* = min\{UB^*, UB^{\varrho}\}$, the best lower bound $LB^* = max\{LB^*, LB^{\varrho}\}$. Compute the current gap $\eta = UB^* - LB^*$. Then find the violated constraints and update the related multipliers according to section 5.2.1 and formula (34). Put the violated constraints into the constraints pool.

end while

$$\begin{cases} \min \sum_{e\in E}\left(\sum_{e^t\in R_e} r_{e^t} - P_e\right) \\ s.t. \quad C_{mpc}, (c_\theta), (c_\lambda) \\ \forall e \in E, \quad r_{e^t} \in N, \quad \lambda_{e^t} \in 0,1 \\ \forall q \in A, \quad \tau_q, \gamma_q, \zeta_q \in 0,1 \end{cases} \tag{33}$$

The constraints in model ($\aleph_f$) consist of three parts. The first part is constraints (C_{mpc}), which is the union of constraints set C_{mpc} in model ($\aleph_u$) and ($\aleph_d$). ζ_a other than those related to constraints (17) and (21) are fixed to their values obtained by Eq. (31), which construct constraints (c_θ). The last part is constraints (c_λ), in these constraints, the platform assignment of each train is fixed as the result of this iteration. Model (33) is a timetabling model with fixed platform assignment decisions.

$$\begin{aligned} \theta_v^{1,k+1} &= \theta_v^{1,k} + \rho \times (\zeta_{q_1,k} - \zeta_{q_2,k} - 1) \\ \theta_v^{2,k+1} &= \theta_v^{2,k} + \rho \times (\zeta_{q_1,k} - \zeta_{q_2,k}) \end{aligned} \tag{34}$$

UB^k and LB^k are the original problem's upper bound and the relaxed problem's lower bound in the kth iteration. The maximum computation time and the gap between the upper and lower bounds are used as the termination criteria. The iterative process can be terminated if one of the two is satisfied.

6. Case study

In this section, to demonstrate the effectiveness of the proposed joint rescheduling model and solution method, several numerical experiments are conducted based on the actual Beijing–Shanghai HSR scheduled timetable. All experiments are implemented in Python 3.6 and executed on a computer with an Intel(R) Core(TM) i5-6500 processor CPU 3.20 GHz and 16 GB memory. The MILP models are solved by commercial solver IBM ILOG CPLEX 12.9.0.

In Section 6.1, the basic data are introduced, and the value of some parameters is determined. In Section 6.2, two types of comparative experiments are executed: the comparison with other strategies, and the comparison with rescheduling in case of the predefined perturbations.

6.1. Basic information

In this paper, the data in the experiments are based on the real timetable and line infrastructure structure of the Beijing–Shanghai HSR. The upstream and downstream are both considered. The Beijing–Shanghai HSR has 23 stations, which start from Beijing South

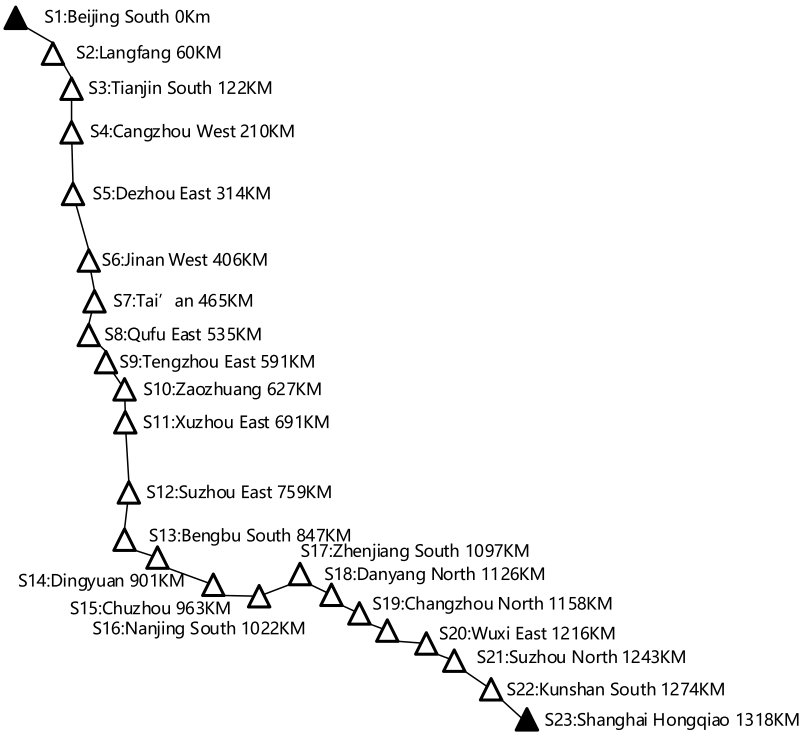


Fig. 10. The layout of Beijing–Shanghai HSR.

Table 4
The minimum running time of each type of train in each section.

Section	Low (min)	High (min)
S1–S2	16	15
S2–S3	12	11
S3–S4	18	16
S4–S5	21	18
S5–S6	19	17
S6–S7	12	10
S7–S8	14	12
S8–S9	11	10
S9–S10	7	6
S10–S11	13	11
S11–S12	14	12
S12–S13	18	15
S13–S14	11	10
S14–S15	12	11
S15–S16	13	13
S16–S17	14	13
S17–S18	6	5
S18–S19	6	6
S19–S20	11	10
S20–S21	5	5
S21–S22	6	6
S22–S23	12	10

Station and end at Shanghai Hongqiao Station. Its mileage is 1318 km. In the daily scheduled timetable of Beijing–Shanghai HSR, trains can be divided into two types. One has a maximum speed of 350 km/h, and the other has a maximum speed of 300 km/h. The layout of the Beijing–Shanghai HSR and the index of each station are shown in Fig. 10, and the minimum running time in each section is given in Table 4.

The computational experiments include six perturbations to test the proposed model's performance in different scenarios, as shown in Table 5. In each perturbation, 50 scenarios are randomly generated by the interval length of 0.2 min. The probability of each scene can be calculated according to Section 4.4.

Table 5
The information of the wind.

Index	Wind section	Additional running time (min)	Shape parameter	Start time	End time
1	3	20	20	14:00	15:00
2	5	15	15	7:30	8:30
3	6	20	20	10:00	11:30
4	9	15	15	18:00	20:00
5	10	20	20	10:50	11:50
6	17	10	10	7:30	8:40

Table 6
The average computation time, total delay, and total affected events of each strategy.

Strategy	Average computation time (s)	Maximum computation time (s)	Total delay (min)	Total affected events	Reduced delay		Reduced affected events	
					P_c^d	P_s^d	P_c^e	P_s^e
ADMM-J	78.80	300	34 585	3862	41.00	39.18	6.73	7.45
S-Joint	38.96	360	41 419	3686	29.34	27.17	10.99	11.67
B-FCFS	39.32	360	47 866	4233	18.35	16.84	2.22	1.43
S-FCFS	10.72	48.52	57 325	4141	0.00	−0.79	0.00	0.76
B-FSFS	37.21	360	45 673	4247	22.09	19.69	−2.55	−1.77
S-FSFS	1.59	5.72	56 873	4173	2.98	0.00	−0.77	0.00
Un-Joint	3.76	21.41	42 491	3911	27.53	25.28	5.55	6.27

Besides, the confidence level α of constraints (26) is set to be 0.8 in 6.2.1. The gap in each iteration of ADMM is 1. The control and predictive horizon length are 20 min and 40 min, respectively. In ADMM-J, the tolerance gap of CPLEX is set to 0.05.

6.2. Computational experiments and comparison of results

Two experiments are carried out to demonstrate the effectiveness of the proposed method. Firstly, in Section 6.2.1, the proposed method is compared to other strategies. Then, it is compared with rescheduling in the case of predefined perturbation in Section 6.2.2. In each experiment, the solution's framework is MPC, shown in Section 5.1. Furthermore, unless otherwise specified, the solution of the model at each stage of MPC is directly solved by CPLEX.

6.2.1. The comparison between the proposed method and other strategies

In this section, experiments with different strategies in case of uncertain perturbations are carried out to demonstrate the effectiveness of the proposed strategy. Firstly, the proposed method is compared with five strategies: S-FCFS, B-FCFS, S-FSFS, B-FSFS, and S-Joint. Furthermore, this section compares it with the train timetable rescheduling without considering platforming (un-joint method). The un-joint method is represented by "Un-Joint" in the following. FCFS denotes first-come-first-served, and FSFS denotes first-scheduled-first-served. S represents that the related strategy does not allow trains to select the reverse routes and platforms, while B is the opposite. In S-Joint, the timetable and platform assignment are jointly optimized. However, it does not allow trains to select reverse tracks. Then, the real-time performance of ADMM-J is investigated by comparing it with the method Joint, in which CPLEX directly solves the model of the prediction horizon. In S-joint, S-FCFS, B-FCFS, S-FSFS, B-FSFS, and Un-joint. The maximum computation of CPLEX is set to be 360 s.

As shown in Table 6, each strategy's average computation time, maximum computation time, total delay, and total affected events are listed. Fig. 17 shows the total delay in the two directions. If an event occurs differently from its scheduled time, it is considered an affected event. What needs to be emphasized is that arrival and departure events are included in calculating the total affected events. P_c^d and P_s^d are the ratio of reduced delays compared with S-FCFS and S-FSFS. P_c^e and P_s^e are the ratio of reduced affected events compared with S-FCFS and S-FSFS.

The proposed method, ADMM-J, has the minimum delay compared to others. Its total delay is 34 585 min. Compared with S-FCFS and S-FSFS, it decreases delay by 41.00% and 39.18%. Compared with B-FCFS and B-FSFS, it decreases delay by 27.74% and 24.27%. These four percentages mean that the proposed method has a high-quality result and dramatically reduces total delay compared with the commonly used rescheduling strategies.

Figs. 11 and 12 show the total delay and the number of total affected events in the two directions, respectively. The light green refers to the downstream, and the light pink represents the upstream. Regardless of the method, the total downstream delay is always larger than the upstream. However, the number of affected events in downstream is always less than the number of affected events in upstream. The average delay in downstream is larger.

Further analysis showed that the proposed method is also less influential. As shown in Table 6, sort according to the number of affected events, S-Joint < ADMM-J < Un-Joint < S-FCFS < S-FSFS < B-FCFS < B-FSFS can be gained. S-Joint has 3686 affected events and 176 events less than ADMM-J. However, due to the less total delay that ADMM-J has, the average delay of ADMM-J is almost 2.3 min less S-Joint.

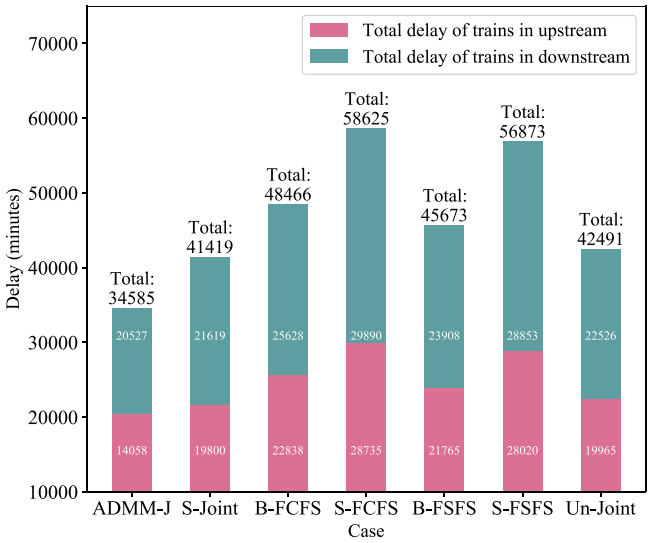


Fig. 11. The total delay of each strategy.

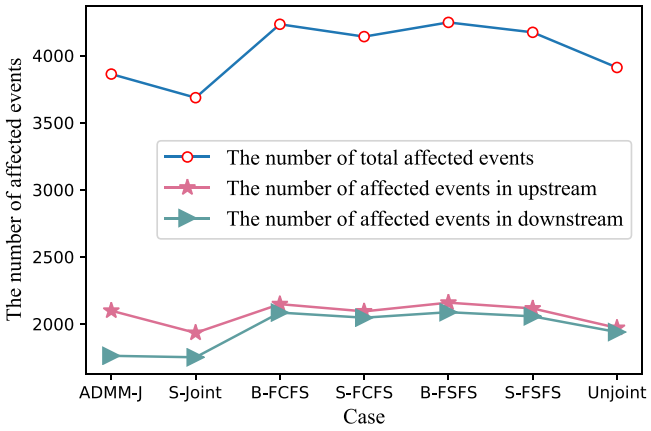


Fig. 12. The number of total affected events of each strategy.

The second column in Table 6 shows that the computation time of the proposed method is the longest. The average computation times of S-FSFS and S-FCFS are less than 11 s. At most, the maximum computation times of each prediction horizon in the above two are smaller than the 50 s. However, in B-FSFS and B-FCFS, the average time and maximum computation time are relatively large. The computation time of some prediction horizons has reached the maximum computation time. In B-FSFS and B-FCFS, since trains in different directions are considered simultaneously, allocating the platforms is quite tricky. However, the departure order between trains in the same direction is fixed. Though the above four strategies can get acceptable results quickly, the quality of these results is relatively low. They waste quantitative railway infrastructures to resume the scheduled operation.

As shown in Fig. 13, the average computation time of the proposed method is in the 80 s. The computation time of most of them is distributed between 1 s and 170 s. To verify that the dualized constraints and ADMM algorithm increase the calculation efficiency, we compare the computation time of ADMM-J and Joint at each prediction horizon, as shown in Fig. 14. In Joint, the computation time of the eight prediction horizons reaches the 1200 s. ADMM-J decreases the computation time by an average of 307.96 s and 79.37%. Only the computation times of the proposed method in the stages located at the beginning and end part of the timetable are greater than the times of the method Joint. The reason is that in the stages at the beginning and end of the timetable, trains rarely encounter trains traveling in the opposite direction. Thus, the number of constraints on trains in different is relatively few. These constraints only increase a little burden on computation. The several iterative computations in ADMM are more time-consuming than solving directly with CPLEX in these stages. In the whole process of MPC, the total computation time (the sum of the computation time of all prediction horizons in MPC) of ADMM-J and Joint is 63.13 and 303.9 min, respectively. The total computation time above is the sum of the computation time of each prediction horizon in MPC.

The total delay and number of affected events in Joint are 42 785 min and 3729. Therefore, the proposed solution can get higher quality results in less time. As shown in Table 6 and Fig. 13, the average computation time of the prediction horizon in ADMM-J is

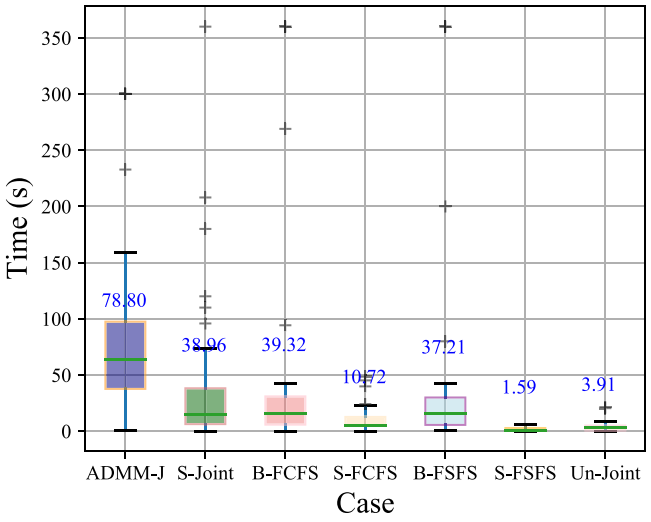


Fig. 13. The real-time performance of each strategy.

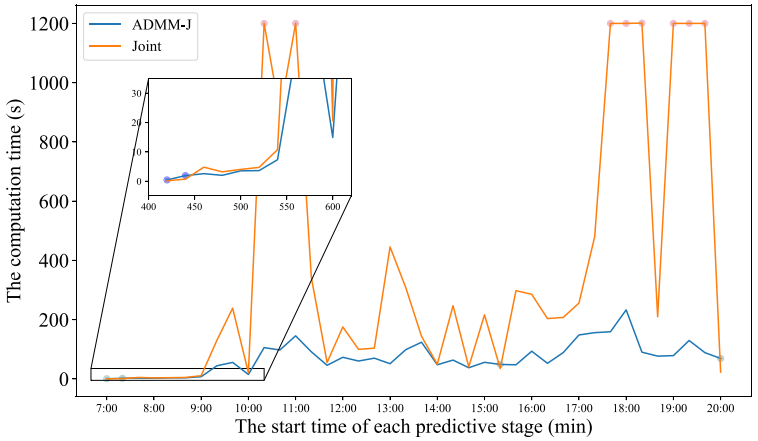


Fig. 14. The computation time of each stage.

78.80 s. Within the framework of MPC, outputting the rescheduling result of a control horizon before it starts is feasible in the real world. Therefore, the proposed method can deal with real-time train rescheduling ideally. However, the state of the train operation environment can change during rescheduling. For example, the perturbation occurs or terminates. In these scenarios, the proposed method seems cannot satisfy the real-time performance. Nevertheless, it is still available by changing part of it. When the state changes, a new stage starts and triggers rescheduling. Here, the model in the prediction horizon is changed to the model in FCFS-S or FSFS-S to get a feasible solution in a short time. Then, the models in the following prediction horizons are kept. Therefore, in this way, before the rescheduled plan should be executed, the rescheduled plan can always be outputted by the proposed method.

Table 7 shows results about allowing trains to select the reverse platforms in the stations. The symbol P denotes the reduced delay compared to the current strategy without allowing trains to use the reverse platform. In ADMM-J, 453 selecting reverse platform events take place. These events reduce 6834 min of delay, which accounts for 16.49% of S-Joint. B-FCFS reduces 10 159 min delay compared to S-FCFS, and B-FSFS reduces 11 200 min delay compared to S-FSFS. In B-FCFS, 188 selecting the reverse platform events take place. They decrease delay by 18.35% and the average delay from 13.84 to 11.37 min. B-FSFS also has a similar result. It decreases by almost 2.87 min on average delay. Hence, even when the departure order is not optimized, allowing trains to occupy the reverse platforms in the station is still an effective way to promote the carrying capacity of railways and decrease the total delay. It helps the original strategy to strengthen the recovery ability.

To test the effectiveness of the joint method, we experiment in case of uncertain perturbation to compare the proposed method with the non-joint method. Here, we consider the non-joint method as train timetable rescheduling without considering platforming. In Un-Joint, the arrival and departure times can be adjusted. Moreover, the departure orders between trains can be flexible. Nevertheless, the stop pattern of each train is fixed, which means that additional dwell is forbidden. Besides, trains must select the

Table 7
The effectiveness of allowing trains to occupy the reverse platforms.

Strategy	Reduced delay P	The number of trains Selecting reverse platforms
ADMM-J	16.4%	453
B-FCFS	16.5%	199
B-FSFS	19.7%	176

Table 8
The comparison between joint and non-joint method.

Method	The number of overtaking	The number of additional dwell	The number of changed platform selection
ADMM-J	379	85	500
Un-Joint	237	0	0

scheduled platforms to dwell or pass through. Correspondingly, the non-joint method does not allow trains to occupy the reverse routes and platforms.

As shown in Table 8, the number of overtaking, changed station activity, and platform selection is listed. ADMM-J changes 85 station activities, which is also the number of additional dwell increases the average dwell time by 1.91 min. The total delay of the Un-Joint method is 42 491 min. The joint method decreases the total delay at 18.6% by changing 500 platform selections and completing 453 utilizing reverse platforms. Due to the stop pattern and platform selection of each train being fixed, the solution space in Un-Joint is significantly reduced, the average computation time of the un-joint method is only 3.76 s, and the maximum computation time is less than 22 s. Comparing S-Joint and Un-Joint, the difference is that ADMM-J can add additional dwell and change the planned track. S-Joint decreases a small amount of delay compared with Un-Joint. The joint method has higher solution quality, regardless of whether trains selecting reverse tracks is allowed.

6.2.2. The comparison between the uncertain and predefined perturbations

In order to demonstrate that the proposed method can deal with uncertain perturbations, a comparison is made between the proposed rescheduling method and a method with predefined perturbations (ADMM-J-Pre). Predefined perturbations refer to the setting of additional running times caused by perturbations in each prediction horizon of MPC to be the actual values in the last control horizon. The main difference between ADMM-J and ADMM-J-Pre lies in the chance constraints about the uncertain impact of perturbations. In ADMM-J-Pre, the chance constraints are substituted by constraints about a predefined scenario. The computation parameters are identical in each prediction horizon of ADMM-J or ADMM-J-Pre.

In Fig. 15, the process of comparing the ADMM-J and ADMM-J-Pre is shown. The first step is initializing the start time T_0, also the start time of the first control and prediction horizon, followed by the main loop. After generating the control and prediction horizon, the event belonging to the prediction horizon is searched for, and go to Step S_4 to reschedule the prediction horizon. If at least one perturbation exists in this stage, then a non-empty set of perturbations Γ_h can be gained. If the set Γ_h is not $\emptyset$, then go to Stop S_5. In Step S_5, for each $p \in \Gamma_h$, let p generate a real scenario p_s to represent the real impact of p in control horizon $[h, h + h_c]$. Then, according to several real scenarios, the rescheduling results within the control horizon can be tested, and the number of conflicts can be counted. If there are conflicts, the rescheduling result is infeasible and triggers rescheduling the control horizon again based on the actual scenario. The control action can be executed only when no conflicts exist in the control horizon. At the end of a full day of operation, the rescheduled result can be outputted, consisting of rescheduled timetable, platform assignment decisions, the number of conflicts, and the rescheduling times.

Step S_5 makes up the main loop of testing robustness, emphasized by the yellow box in Fig.fig:robustness test. What is to be emphasized is that Step S_4 is different in ADMM-J and ADMM-J-Pre. In ADMM-J, the additional running times of perturbations are still unknown and constructed as chance constraints. However, the perturbations are assumed to be the real scenarios generated in the last control horizon in ADMM-J. The real additional times in the last loop are used to construct the deterministic constraints about virtual stations in ADMM-J-Pre to substitute the chance constraints of ADMM-J. It means that the impact of perturbations in the prediction horizon of a stage can only be determined and renewed after rescheduling this stage.

In order to test the performance of the proposed method, three situations are designed. The information about the perturbations of each situation is shown in Table 9. According to the (28), the wind speed distribution range is relatively large if the perturbation has a relatively small shape parameter. It means that its uncertainty is relatively high. Therefore, the three situations are named high, medium, and low uncertainty according to the shape parameters. In each situation, we compare the ADMM-J with a high confidence level (0.7), ADMM-J with a medium confidence level (0.5), ADMM-J with a low confidence level (0.3), and ADMM-J with predefined perturbations together. In the following content, ADMM-J-0.7, ADMM-J-0.5, ADMM-J-0.3, and ADMM-J-pre denote the above four ADMM-J.

The robustness of a rescheduling result indicates its ability to remain feasible even when the states of perturbations change. Two evaluation indexes are employed to evaluate the robustness of the rescheduling method. The first is the total conflicts. A result without conflicts is more robust when the perturbations change, but a result with fewer conflicts may not necessarily be more robust. For example, as shown in Fig. 15, here we suppose there are two rescheduling results in the main loop. If the result a has ten conflicts

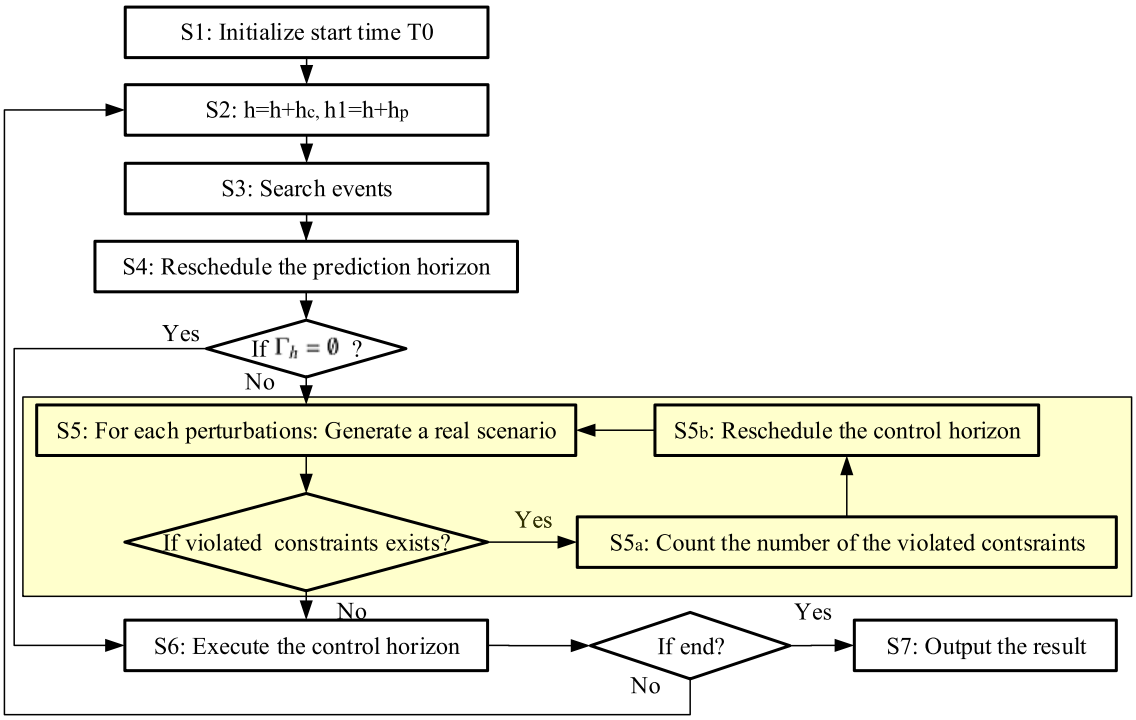


Fig. 15. The process of testing the robustness of the rescheduling method.

Table 9
The information about perturbations of each situations.

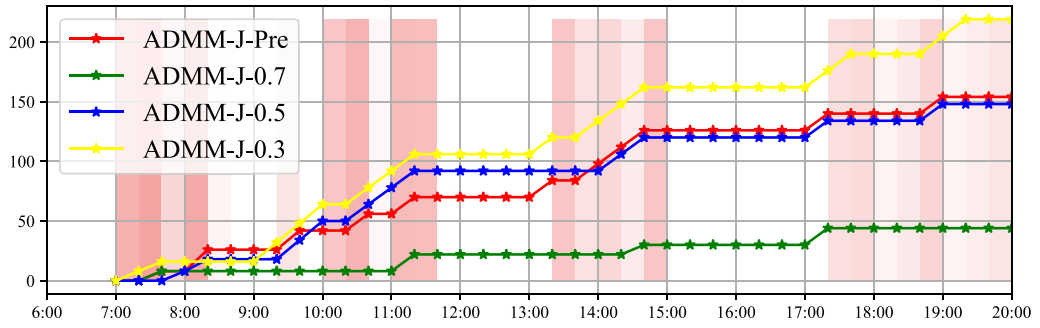
Index	Additional running time (min)	High Shape parameter	Medium Shape parameter	Low Shape parameter
1	20	8	20	40
2	10	4	10	20
3	20	8	20	40
4	20	8	20	40
5	20	8	20	40
6	20	8	20	40

and the result *b* has 20 conflicts, both *a* and *b* cannot be executed. Rescheduling with the renewed information of perturbations has to be triggered. Hence, a second index is needed: the number of rescheduling times. In a stage, if result *a* has no conflict, and result *b* has conflicts, then *a* can be executed directly, while *b* needs to be rescheduled. Hence, *a* only triggers one rescheduling, and *b* triggers two times. *a* has better robustness and takes less computation time.

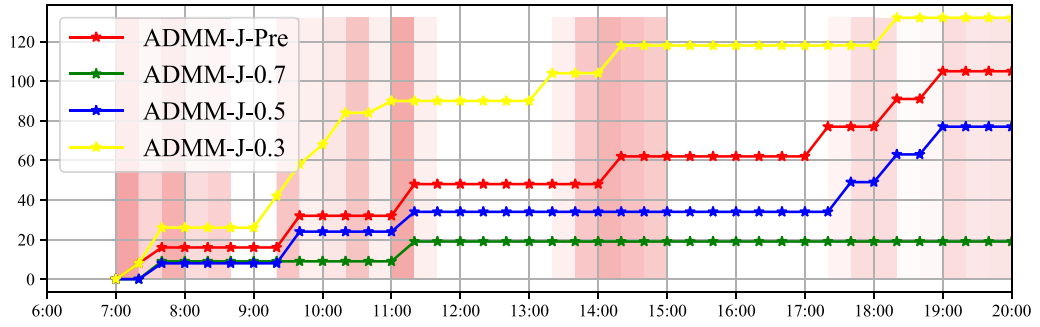
The number of conflicts in the three situations is shown in Figs. 16(a), 16(b), and 16(c), respectively. The lines represent the cumulative value of the number of conflicts. If the line goes up, new conflicts exist in the current control horizon, and a rescheduling with this horizon should be triggered. The background color in these three figures denotes the degree of difference in actual running times between the current control horizon and the last control horizon. The darker the background, the greater the difference. ADMM-J-0.7 and ADMM-J-0.5 have fewer conflicts than ADMM-J-Pre in all situations. Their rescheduling results are relatively stable and feasible than the other two methods. The ADMM-J-0.7 only triggers rescheduling four times due to the infeasibility within the control horizon in case of high uncertainty and two and one times in the situation with medium and low uncertainty, respectively. In high and medium uncertainty, ADMM-J-0.3 is worse than ADMM-J-Pre. Nevertheless, ADMM-J-0.3 is slightly better than ADMM-J-Pre in case of low uncertainty.

Fig. 17 displays each case's total delay and rescheduling times. The total delay and rescheduling times are marked at each colored block. The colored blocks indicate the number of reschedulings, with colors closer to green representing greater rescheduling and those closer to yellow indicating fewer reschedulings. A lower total rescheduling count indicates less computation time, better feasibility, and applicability.

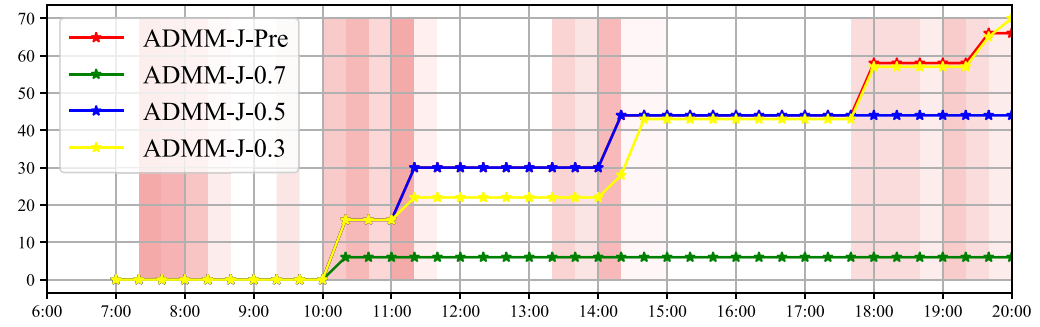
In high-uncertainty scenarios, ADMM-J-0.7 had the fewest reschedulings, at 53, and reduced delays by 402 min compared to ADMM-J-Pre, with conflicts in 4 control horizons. It demonstrated the highest level of robustness. ADMM-J-0.5 and ADMM-J-Pre had the same total rescheduling times and similar robustness levels, but ADMM-J-0.5 had 1207 min less delay than ADMM-J-Pre. The robustness of the two is also close. However, ADMM-J-0.5 has 1207 min delay less than ADMM-J-Pre. ADMM-J-0.3 had the



(a) The number of conflicts in case of high uncertainty.



(b) The number of conflicts result in case of medium uncertainty.



(c) The number of conflicts result in case of low uncertainty.

Fig. 16. The cumulative number of conflicts.

least robustness despite a minor delay, with four times rescheduling more than ADMM-J-Pre and the most conflicts. Therefore, ADMM-J-0.7 is more suitable in highly uncertain situations as it is less time-consuming and more robust. Furthermore, ADMM-J-0.5 find a compromise between robustness and quality.

Compared to ADMM-J-Pre, methods that considered uncertainty reduced delays by an average of 1217 min compared to the situation with medium uncertainty. ADMM-J-0.7 and ADMM-J-0.5 have better robustness than the other two methods. They decrease total rescheduling times by six and two times compared to ADMM-J-Pre. The results in the situation with low uncertainty are similar. ADMM-J-0.7 and ADMM-J-0.5 have better results and robustness than the other two methods. They improve the quality of the adjustment strategy quality while promoting its robustness.

The ADMM-J-0.7 is the most robust in all situations, with relatively worse quality. It has to sacrifice an inevitable delay to promote its robustness. However, ADMM-J-0.3 gets minor delays. It always uses a low confidence level to make trains pass through the sections where perturbations occur in as little time as possible. If the result is feasible in the control horizon with the

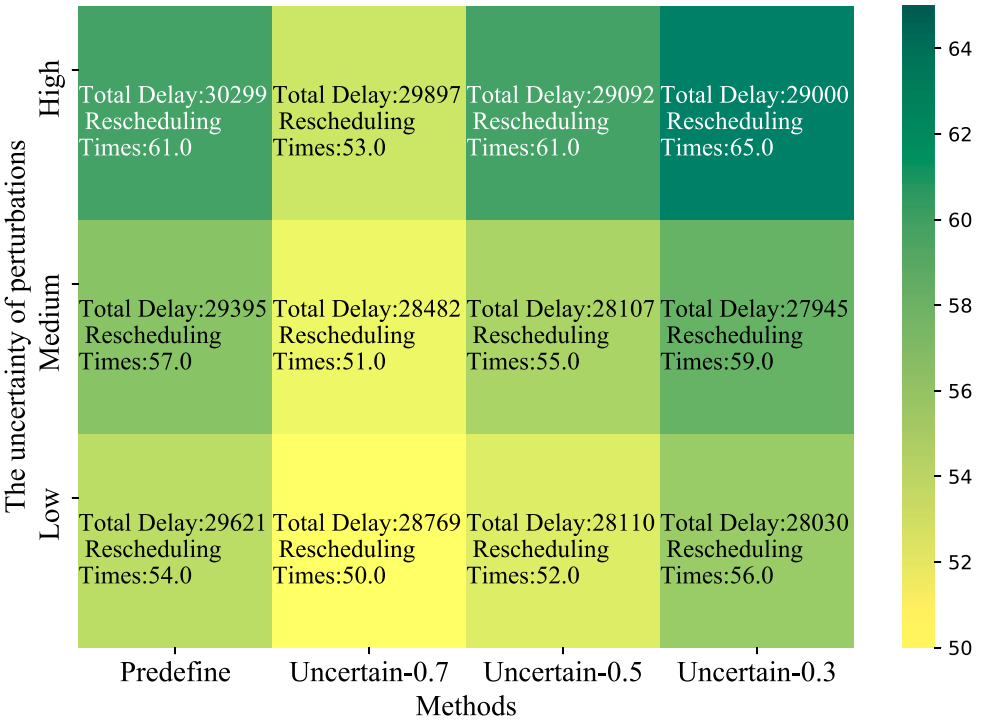


Fig. 17. The total delay and rescheduling times of each case.

actual scenario, then the result in this stage contributes less delay in the entire rescheduling result. The proposed model with low uncertainty always assesses the disturbance uncertainty more positively. Its behavior is a high-risk, high-reward behavior.

7. Conclusion and future research

This paper proposes a joint rescheduling method for high-speed railway timetabling and platforming in case of uncertain perturbations. The proposed method includes retiming, reordering, rerouting, and additional dwell. It also allows trains to occupy the reverse routes and platforms. A mixed-integer linear programming model with chance constraints is established based on the event-activity network. MPC is utilized to divide the large-scale model into several stages. The constraints about trains in different directions are dualized, which decreases numerous constraints and the computation burden of the proposed model. The ADMM algorithm can solve the dualized problem in parallel.

The practical scheduled timetable and line data of Beijing–Shanghai HSR are utilized in the numerical experiments. Computational experiments show that compared to the commonly used strategy, the proposed model can reduce the total delay by more than 39%. It reduces the delay by 18.6% by reasonably adding additional dwell and adjusting platform assignment, compared to the method that only includes timetabling. Besides, it shows better robustness than rescheduling in case of predefined perturbations and has less delay, times of rescheduling, and the number of conflicts.

Further research is needed to consider rescheduling rolling stock for long-duration disruption and more flexible interlocking mechanisms. The joint optimization method for timetable and platform assignment with passenger demand will also be explored.

CRediT authorship contribution statement

Xuan Liu: Conceptualization, Methodology, Investigation, Writing – original draft, Writing – review & editing. **Min Zhou:** Data curation, Writing – original draft. **Hairong Dong:** Validation, Investigation, Writing – review & editing. **Xingtang Wu:** Software, Visualization. **Yidong Li:** Validation, Investigation. **Fei-Yue Wang:** Conceptualization, Resources, Supervision.

Acknowledgments

The research in the paper was jointly supported by the National Natural Science Foundation of China (No. 61925302, No. 62103033) and by the Technological Research and Development Program of China Railway Corporation (P2022X013).

Table 10
The value of $\left[\zeta_{q_1}+\zeta_{q_2}-1\right]^2$ and $\left[\zeta_{q_1}-\zeta_{q_2}\right]^2$.

$\left[\zeta_{q_1}+\zeta_{q_2}-1\right]^2$	$\left[\zeta_{q_1}-\zeta_{q_2}\right]^2$	ζ_{q_1}	ζ_{q_2}
1	0	0	0
0	1	0	1
0	1	1	0
1	0	1	1

Appendix A. The linearization and analysis of constraints (14)

First, the symbols $\overline{\kappa^{i,t}}=\lambda_{e_{i,t}}\times\kappa^{t}_{out}\times(1-\zeta_q)$ if the headway activity $q\in A^{H}_{dep}$. Otherwise, $\overline{\kappa^{i,t}}=-\lambda_{e_{j,t}}\times\kappa^{t}_{in}\times\zeta_q$. $\lambda_{e^{i,t}}\times\zeta_q$ is nonlinear. A new type of decision variable, ς^0, is introduced to substitute the non-linear term. Only both $\lambda_{e^{i,t}}$ and ζ_q are 1, the $\varsigma^0=1$. Otherwise $\zeta_q=0$. The relationship between them is shown in

$$\begin{cases}\varsigma^0\le\lambda_{e^{i,t}}\\ \varsigma^0\le\zeta_q\\ \varsigma^0\ge\lambda_{e^{i,t}}+\zeta_q-1\end{cases}\tag{35}$$

Then is the analysis of the constraints (14). Here, the activity q is regarded as the departure headway activity. Taking the two trains k and j in Fig. 7 as an example. The constraints (14) regarding to trains k and j with the departure headway q can be written as the following:

$$\begin{aligned}\omega_q+r_{e^{j,3}}+\kappa^{3}_{out}\times\zeta_q\le r_{e^{k,6}}+\kappa^{6}_{out}\times(1-\zeta_q)+(1-\zeta_q)\times M\\ \omega_q+r_{e^{k,6}}+\kappa^{6}_{out}\times(1-\zeta_q)\le r_{e^{j,3}}+\kappa^{3}_{out}\times\zeta_q+\zeta_q\times M\end{aligned}\tag{36}$$

If $\zeta_q=1$, train j departs earlier than train k, as shown in Fig. 7. And constraint (36) is converted to $\omega_q+r_{e^{j,3}}+\kappa^{3}_{out}\le r_{e^{k,6}}$. It means that train k can only depart from the platform and occupy only after train j releases the outbound route 6 for a while. If $\zeta_q=0$, $\omega_q+r_{e^{k,6}}+\kappa^{6}_{out}\le r_{e^{j,3}}$ can be gained. Train j should occupy its outbound route only after train k releases its outbound for a while.

$$\begin{aligned}\omega_q+r_{e^{j,3}}-\kappa^{3}_{in}\times(1-\zeta_q)\le r_{e^{k,6}}-\kappa^{6}_{in}\times\zeta_q+(1-\zeta_q)\times M\\ \omega_q+r_{e^{k,6}}-\kappa^{6}_{in}\times\zeta_q\le r_{e^{j,3}}-\kappa^{3}_{in}\times(1-\zeta_q)+\zeta_q\times M\end{aligned}\tag{37}$$

Constraint (37) represents that q is an arrival headway activity. If $\zeta_q=1$, $\omega_q+r_{e^{j,3}}\le r_{e^{k,6}}-\kappa^{6}_{in}$. Otherwise, $\omega_q+r_{e^{k,6}}\le r_{e^{j,3}}-\kappa^{3}_{in}$. A safe arrival headway can be guaranteed.

The constraints (20) can be linearized and analyzed similarly.

Appendix B. The linearization process of constraints (17)

The quadratic term $\left[\zeta_{q_1}+\zeta_{q_2}-1\right]^2$ in Eq. (31) is derived from constraints (17). Only three possible results for $\zeta_{q_1}+\zeta_{q_2}-1$ can be gained, which are $-1,0,1$. Hence, $\left[\zeta_{q_1}+\zeta_{q_2}-1\right]^2$ can only be equal to 1 or 0, as shown in Table 10. The value of $\left[\zeta_{q_1}+\zeta_{q_2}-1\right]^2$ can be deduced from the logical relationship between ζ_{q_1} and ζ_{q_2}, shown as the following table. $\forall(q_1,q_2)=\nu\in Y^{o}_{r}$, if $\zeta_{q_1}+\zeta_{q_2}=1$, $\left[\zeta_{q_1}+\zeta_{q_2}-1\right]^2=0$, otherwise $\left[\zeta_{q_1}+\zeta_{q_2}-1\right]^2=1$. Here, the decision variables ς^1 are introduced, and it is utilized to denote whether only one of ζ_{q_1} and ζ_{q_2} is equal to 1. The relationship between them can be modeled as the following linear constraints:

$$\begin{cases}\varsigma^1\ge\zeta_{q_1}-\zeta_{q_2}\\ \varsigma^1\ge\zeta_{q_2}-\zeta_{q_1}\\ \varsigma^1\le\zeta_{q_1}+\zeta_{q_1}\\ \varsigma^1\le2-\zeta_{q_1}-\zeta_{q_1}\end{cases}\tag{38}$$

Appendix C. The linearization process of constraints (21)

The linearization process of the quadratic term $\left[\zeta_{q_1}-\zeta_{q_2}\right]^2$ is similar to the above process. $\forall(q_1,q_2)=\nu\in Y_{rr}$, only when $\zeta_{q_1}=\zeta_{q_2}$, $\left[\zeta_{q_1}-\zeta_{q_2}\right]^2=0$, otherwise it will be equal to 1, as shown in Table 10. Similarly, the decision variables ς^2 are introduced. It is used to judge whether $\zeta_{q_1}=\zeta_{q_2}$ according to the following linear constraints:

$$\begin{cases}\varsigma^2\le\zeta_{q_1}\\ \varsigma^2\le\zeta_{q_2}\\ \varsigma^2\ge\zeta_{q_1}+\zeta_{q_1}-1\end{cases}\tag{39}$$

References

Ata, B., Ml, C., Uw, A., 2019. Coordinated railway traffic rescheduling with the resource conflict graph model. J. Rail Transp. Plan. Manage. 15.

Binder, S., Maknoon, Y., Bierlaire, M., 2017. The multi-objective railway timetable rescheduling problem. Transp. Res. C 78.

Cacchiani, V., Galli, L., Toth, P., 2015. A tutorial on non-periodic train timetabling and platforming problems. Euro J. Transp. Logist. 4 (3), 285–320.

Cacchiani, V., Huisman, D., Kidd, M., Kroon, L., Toth, P., Veelenturf, L., Wagenaar, J., 2014. An overview of recovery models and algorithms for real-time railway rescheduling. Transp. Res. B 631 (May), 15–37.

Cacchiani, V., Qi, J., Yang, L., 2020. Robust optimization models for integrated train stopping planning and timetabling with passenger demand uncertainty. Transp. Res. B 136, 1–29.

Cadarso, L., Marín, Ángel, 2014. Improving robustness of rolling stock circulations in rapid transit networks. Comput. Oper. Res. 51, 146–159.

Cavone, G., van den Boom, T., Blenkers, L., Dotoli, M., Seatzu, C., De Schutter, B., 2020. An mpc-based rescheduling algorithm for disruptions and disturbances in large-scale railway networks. IEEE Trans. Autom. Sci. Eng. 1–14.

Corman, F., D'Ariano, A., Pacciarelli, D., Pranzo, M., 2010. A tabu search algorithm for rerouting trains during rail operations. Transp. Res. B 44 (1), 175–192.

D'Ariano, A., 2009. Innovative decision support system for railway traffic control. IEEE Intell. Transp. Syst. Mag. 1 (4), 8–16.

D'Ariano, A., Pacciarelli, D., Samà, M., Corman, F., 2017. Microscopic delay management: Minimizing train delays and passenger travel times during real-time railway traffic control. In: 2017 5th IEEE International Conference on Models and Technologies for Intelligent Transportation Systems (MT-ITS). pp. 309–314.

Dong, H., Liu, X., Zhou, M., Zheng, W., Xun, J., Gao, S., Song, H., Li, Y., Wang, F.-Y., 2021. Integration of train control and online rescheduling for high-speed railways in case of emergencies. IEEE Trans. Comput. Soc. Syst. 1–9.

Jiang, F., Cacchiani, C., Toth, P., 2017. Train timetabling by skip-stop planning in highly congested lines. Transp. Res. B 104, 149–174.

Kersbergen, B., Ton, V., Schutter, B.D., 2016. Distributed model predictive control for railway traffic management. Transp. Res. C 68 (jul.), 462–489.

Krasemann, J.T., 2012. Design of an effective algorithm for fast response to the re-scheduling of railway traffic during disturbances. Transp. Res. C 20 (1), 62–78.

Li, S., De Schutter, B., Yang, L., Gao, Z., 2016. Robust model predictive control for train regulation in underground railway transportation. IEEE Trans. Control Syst. Technol. 24 (3), 1075–1083.

Li, X., Meng, L., Wang, Y., Long, S., Miao, J., 2019. A rescheduling optimization model under temporary speed restriction. In: 2019 IEEE Intelligent Transportation Systems Conference - ITSC.

Lu, G., Ning, J., Liu, X., Nie, Y., 2022. Train platforming and rescheduling with flexible interlocking mechanisms: An aggregate approach. Transp. Res. E 159.

Luan, X., De Schutter, B., Meng, L., Corman, F., 2020. Decomposition and distributed optimization of real-time traffic management for large-scale railway networks. Transp. Res. B 141, 72–97.

Luan, X., Wang, Y., De Schutter, B., Meng, L., Lodewijks, G., Corman, F., 2018a. Integration of real-time traffic management and train control for rail networks - part 1, optimization problems and solution approaches. Transp. Res. B 115 (SEP.), 41–71.

Luan, X., Wang, Y., De Schutter, B., Meng, L., Lodewijks, G., Corman, F., 2018b. Integration of real-time traffic management and train control for rail networks - part 2, extensions towards energy-efficient train operations. Transp. Res. B 115, 72–94.

Mannino, C., Andreas, Makkerud, 2023. Optimal train rescheduling in oslo central station. Omega 116.

Meng, L., Zhou, X., 2014b. Simultaneous train rerouting and rescheduling on an n-track network: A model reformulation with network-based cumulative flow variables. Transp. Res. B 67, 208–234.

Meng, L., Zhou, X., 2019. An integrated train service plan optimization model with variable demand: A team-based scheduling approach with dual cost information in a layered network. Transp. Res. B 125 (JUL.), 1–28.

Nielsen, L.K., Kroon, L., Maróti, G., 2012. A rolling horizon approach for disruption management of railway rolling stock. European J. Oper. Res. 220 (2), 496–509, 2012.

Pellegrini, P., Marlière, G., Rodriguez, J., 2014. Optimal train routing and scheduling for managing traffic perturbations in complex junctions. Transp. Res. B 120, 33–48.

Pellegrini, P., Pesenti, R., Rodriguez, J., 2019. Efficient train re-routing and rescheduling: Valid inequalities and reformulation of RECIFE-MILP. Transp. Res. B 59, 58–80.

Peng, Qiyuan, Zhan, Shuguang, Kroon, Leo, G., Zhao, ., Jun, ., 2016. A rolling horizon approach to the high speed train rescheduling problem in case of a partial segment blockage. Transp. Res. E.

Sama, M., D'Ariano, A., Corman, F., Pacciarelli, D., 2017. A variable neighbourhood search for fast train scheduling and routing during disturbed railway traffic situations. Comput. Oper. Res. 78 (feb.), 480–499.

Su, Z., Jamshidi, A., Nunez, A., Baldi, S., De Schutter, B., 2017. Multi-level condition-based maintenance planning for railway infrastructures – a scenario-based chance-constrained approach. Transp. Res. C 84 (nov.), 92–123.

Wang, W., Okaze, T., 2022. Statistical analysis of low-occurrence strong wind speeds at the pedestrian level around a simplified building based on the Weibull distribution. Build. Environ. 209, 108644.

Wang, W., Qin, C., Zhang, J., Wen, C., Xu, G., 2022. Correlation analysis of three-parameter Weibull distribution parameters with wind energy characteristics in a semi-urban environment. Energy Rep. 8, 8480–8498.

Xu, P., Corman, F., Peng, Q., Luan, X., 2017a. A train rescheduling model integrating speed management during disruptions of high-speed traffic under a quasi-moving block system. Transp. Res. B 104, 638–666.

Xu, X., Li, C., Xu, Z., 2017b. Train timetabling with stop-skipping, passenger flow, and platform choice considerations. Transp. Res. B 150, 52–74.

Zhan, S., Kroon, L.G., Veelenturf, L.P., et al., 2015. Real-time high-speed train rescheduling in case of a complete blockage. Transp. Res. B.

Zhan, S., Kroon, L.G., Zhao, J., et al., 2016. A rolling horizon approach to the high speed train rescheduling problem in case of a partial segment blockage. Transp. Res. E 95, 32–61.

Zhan, S., Wong, S., Shang, P., Peng, Q., Xie, J., Lo, S., 2021. Integrated railway timetable rescheduling and dynamic passenger routing during a complete blockage. Transp. Res. B 143, 86–123.

Zhang, Y., D'Ariano, A., He, B., Peng, Q., 2019a. Microscopic optimization model and algorithm for integrating train timetabling and track maintenance task scheduling. Transp. Res. B 127, 237–278.

Zhang, H., Li, S., Wang, Y., Wang, Y., Yang, L., 2021b. Real-time optimization strategy for single-track high-speed train rescheduling with disturbance uncertainties: A scenario-based chance-constrained model predictive control approach. Comput. Oper. Res. 127 (1), 105135.

Zhang, H., Li, S., Wang, Y., Yang, L., Gao, Z., 2020a. Collaborative real-time optimization strategy for train rescheduling and track emergency maintenance of high-speed railway: A lagrangian relaxation-based decomposition algorithm. Omega 102 (5), 102371.

Zhang, Q., Lusby, R.M., Shang, P., Zhu, X., 2020b. Simultaneously re-optimizing timetables and platform schedules under planned track maintenance for a high-speed railway network. Transp. Res. C 121, 102823.

Zhang, Q., Lusby, R.M., Shang, P., Zhu, X., 2022b. A heuristic approach to integrate train timetabling, platforming, and railway network maintenance scheduling decisions. Transp. Res. B 158.

Zhang, Y., Peng, Q., Lu, G., Zhong, Q., Xu, Y., Zhou, X., 2022. Integrated line planning and train timetabling through price-based cross-resolution feedback mechanism. Transp. Res. B 155, 240–277.

Zhang, Y., Peng, Q., Yao, Y., Zhou, X., 2019b. Solving cyclic train timetabling problem through model reformulation: Extended time-space network construct and alternating direction method of multipliers methods. Transp. Res. B 128, 344–379.
Zhu, Y., Goverde, R., 2019. Railway timetable rescheduling with flexible stopping and flexible short-turning during disruptions. Transp. Res. B 123 (MAY), 149–181.
Zhu, Y., Goverde, R.M., 2020. Dynamic and robust timetable rescheduling for uncertain railway disruptions. J. Rail Transp. Plan. Manage. 15, 100196.
Zhu, Y., Goverde, R., 2021. Dynamic railway timetable rescheduling for multiple connected disruptions. Transp. Res. C 125 (3).